

AH CS PROJECT

WDD Integrating DDD

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1. Analysis

1.1 Description of Problem

I intend to create a website which allows individual users to view information about available jobs in their area and businesses to create job postings. For individual users, the website will break available jobs down into categories based on the type of work. Since some of the target end users for my website are teenagers who are applying for their first job, the website will also contain a guide on how to write a CV. For businesses, the website will allow them to create a job posting to be viewed, and applied to, by personal users.

The website will be made using HTML and styled with CSS, additionally I will make use of PHP to make the website dynamic. The website will interact with a database, using SQL, to store information about jobs, employers, and users.

1.2 Constraints

- The project must be completed by the project deadline on the 22nd of March 2024.
- Free, and mostly open-source, software will be used to keep the project free of cost.
- I will make use of Microsoft Office which I can access for free through Glow Scotland.
- The database will be hosted on my local machine, this means there is no associated cost with database hosting.
- As the website will not be published live on the internet, it will only be accessible from my personal machine.
- The user is not applying for a real job, so nothing will happen beyond a message displayed whether they have been successful or not.
- Users cannot apply for the same job multiple times.

1.3 Requirements Specification

1.3.1 End User Requirements

1. A user account can either be an applicant or a business.
2. Applicants and unregistered users must be able to view a CV writing guide.

1.3.1a For Applicants

3. An applicant must be able to sign up for a new account or log into an existing account to save and view available jobs.
4. An applicant must be able to easily navigate between the different pages on the website to view their account information, available jobs, and jobs that they have already applied to.
5. An applicant must be able to view information about each job including job description, contracted hours, expected salary, and employer contact details.
6. An applicant must be able to view their personal information.
7. An applicant must be able to submit a job application to a business.

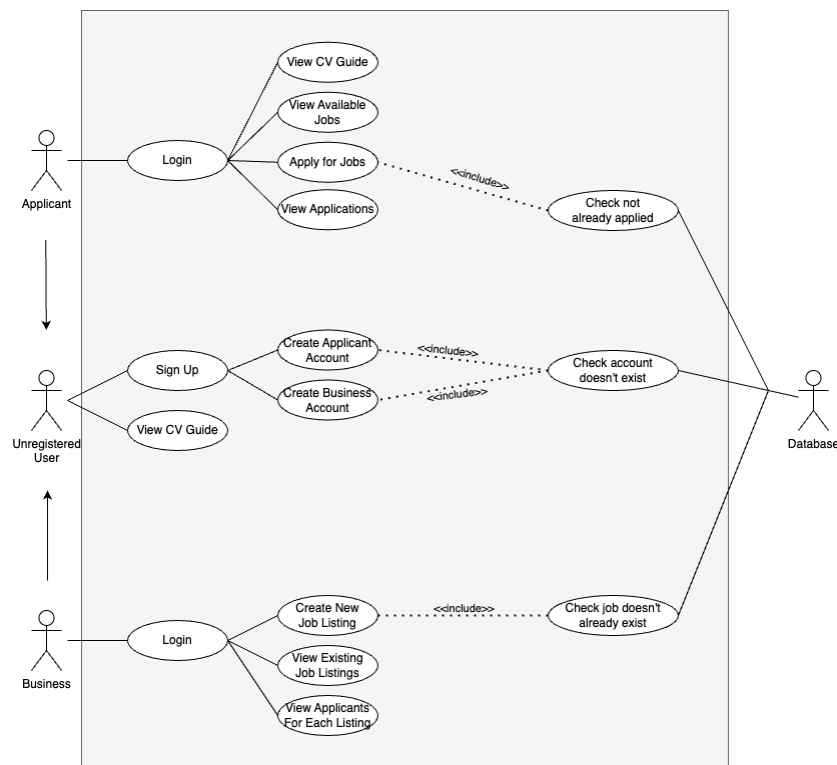
1.3.1b For Businesses

8. Businesses must be able to create a new job listing with details of the job such as a description and hourly rate.
9. Businesses must be able to view their current job listings and view the current applicants of each of their job listings.
10. Businesses must be able to access a business profile to view their information.

1.3.2 Functional Requirements

11. The website must contain a guide on how to write a CV.
12. The website must allow the user to log into an existing account or sign up with a new account.
13. The website must validate all user inputs before storing them in the database.
14. The website must interact with a database to store, access user account information.
15. The website will have a horizontal navigation bar with links to each main page.
16. The website must display information about each job to the applicants including job description, contracted hours, expected salary, and employer contact details.
17. The website must allow business profiles to add a new job. This includes using a form to gather information such as a job title, contracted hours, expected salary and location.
18. The website must have a page with a form which allows all users to edit their account information.
19. The website must show businesses the applicants for each of their job listings.
20. The website should use a media query so when the screen width is less than 600px flex sections are aligned in columns rather than rows.

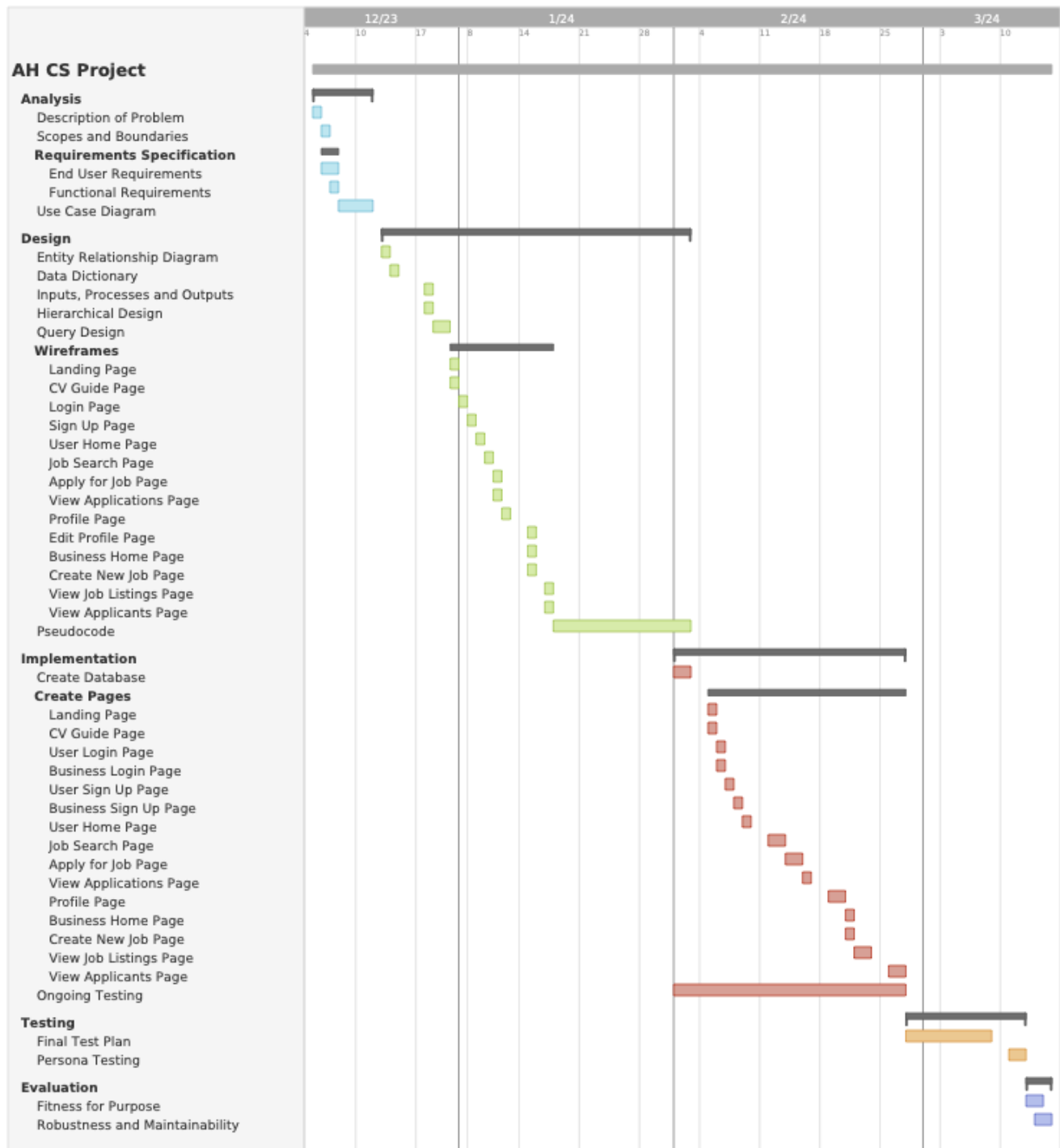
1.4 Use Case Diagram



1.5 Project Plan

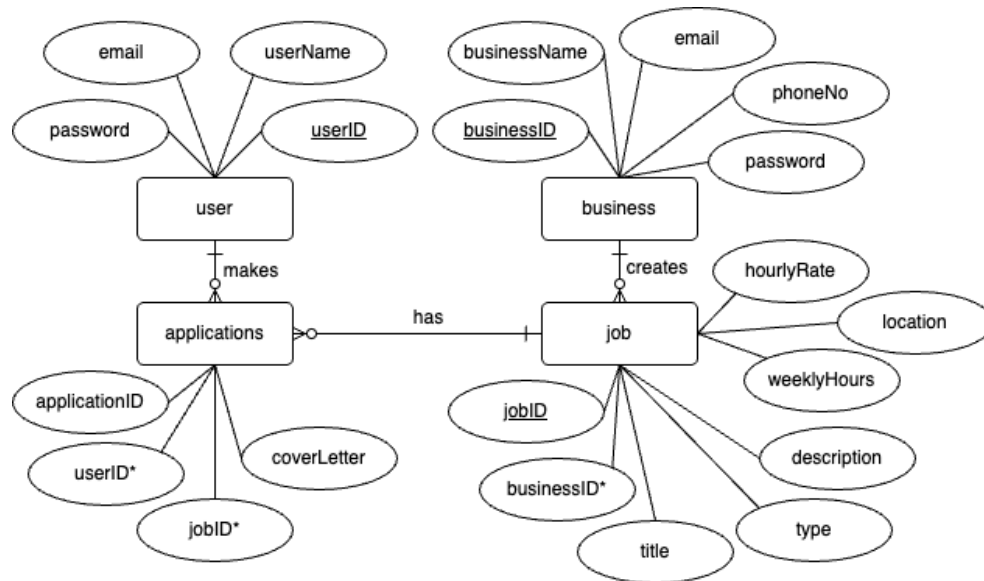
Required resources:

- Visual Studio Code, MySQL Workbench, EasyPHP (Windows) or MAMP (macOS), Google Chrome, Draw.io, Microsoft Word



2. Design

2.1 Entity Relationship Diagram



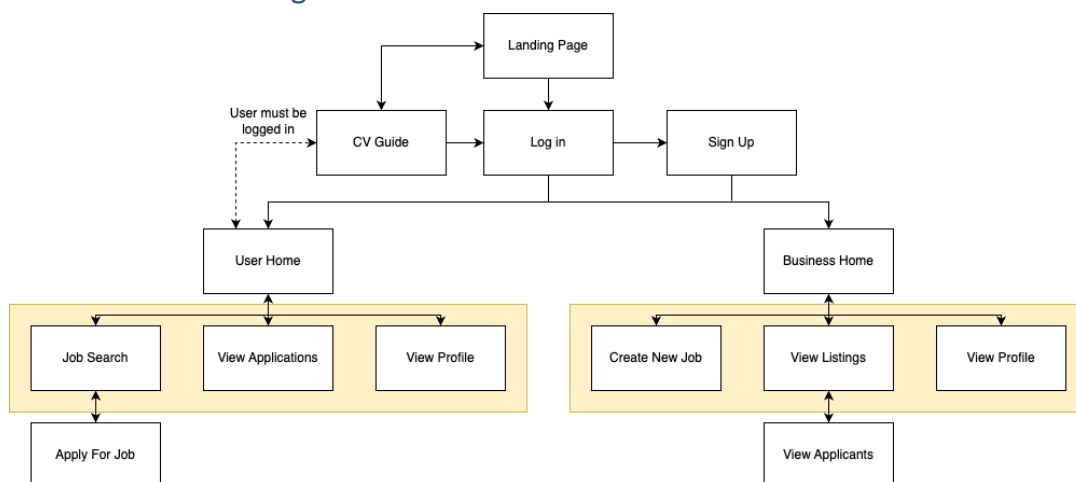
2.2 Data Dictionary

Entity	Attribute	PK/FK	Data Type	Size	Required	Validation
<i>user</i>	userID	PK	Integer		Yes	Auto Increment
	userName		Varchar	70	Yes	
	email		Varchar	100	Yes	
	password		Varchar	30	Yes	Length >= 8
<i>business</i>	businessID	PK	Integer		Yes	Auto Increment
	businessName		Varchar	70	Yes	
	email		Varchar	100	Yes	
	phoneNo		Varchar	15	Yes	
	password		Varchar	30	Yes	Length >= 8
<i>job</i>	jobID	PK	Integer		Yes	Auto Increment
	title		Varchar	80	Yes	
	jobType		Varchar	14	Yes	Limit to 'Full-time', 'Part-time' or 'Apprenticeship'
	businessID	FK	Integer		Yes	Existing businessID from business
	description		Varchar	4000	Yes	
	weeklyHours		Decimal		Yes	Hours <= 168
	hourlyRate		Decimal		Yes	
<i>applications</i>	location		Varchar	200	Yes	
	applicationID	PK	Integer		Yes	Auto Increment
	userID	FK	Integer		Yes	Existing userID from user
	jobID	FK	Integer		Yes	Existing jobID from jobs
	coverLetter		Varchar	2000	No	

2.3 Inputs, Processes and Outputs

Inputs		
When a profile is created: • Name • Email • Password Additionally for businesses: • Phone Number	Businesses creating a job: • Title • Description • Hourly Rate • Weekly Hours • Location	When an application is made: • Cover letter (optional).
Processes		
<i>Note that unique IDs are automatically incrementing integer values.</i>		
When a profile is created: • Validate user input. • Store details in the database. For users: • Generate a unique user ID. For businesses: • Generate a unique business ID.	When a job is created: • Validate job details. • Store job details in the database. • Generate a unique job ID.	When an application is made: • Store application in the database. • Generate unique application ID.
Outputs		
• Display profile details. • Display available jobs. • Display user's current applications. • Display businesses' current job listings.		

2.4 Hierarchical Design



There will be some quick links from the User Home page to the Apply for Job page as well as some quick links from the Business Home page to the View Applicants page.

2.5 Query Design

User log in query.

Fields/Calculations	*
Tables/Queries	user
Search Criteria	user.email = '\$email' AND user.password = '\$pword'
Grouping	
Having	
Sort Order	

Business log in query.

Fields/Calculations	*
Tables/Queries	business
Search Criteria	business.email = '\$email' AND business.password = '\$pword'
Grouping	
Having	
Sort Order	

Check user account does not already exist.

Fields/Calculations	*
Tables/Queries	user
Search Criteria	user.email = '\$email'
Grouping	
Having	
Sort Order	

Create new user account. *(Insert query)*

Fields/Calculations	userName, email, password
Tables/Queries	user
Search Criteria	
Grouping	
Having	
Sort Order	

Check business account does not already exist.

Fields/Calculations	*
Tables/Queries	business
Search Criteria	business.email = '\$email'
Grouping	
Having	
Sort Order	

Create new business account. *(Insert query)*

Fields/Calculations	businessName, email, phoneNo, password
Tables/Queries	business
Search Criteria	
Grouping	
Having	
Sort Order	

User home page display 3 most recent jobs.

Fields/Calculations	*
Tables/Queries	Job
Search Criteria	
Grouping	
Having	
Sort Order	jobID DESC
Limit	3

Profile page for user account.

Fields/Calculations	*
Tables/Queries	user
Search Criteria	user.userID = '\$accountID'
Grouping	
Having	
Sort Order	

Profile page for business account.

Fields/Calculations	*
Tables/Queries	business
Search Criteria	business.businessID = \$accountID'
Grouping	
Having	
Sort Order	

Job search page to search jobs using user's search query.

Fields/Calculations	*
Tables/Queries	job, business
Search Criteria	job.businessID = business.businessID AND jobType = '\$jobType' AND weeklyHours >= '\$wkHours' AND hourlyRate >= '\$hrRate' AND (title LIKE '%\$searchString%' OR location LIKE '%\$searchString%')
Grouping	
Having	
Sort Order	jobID DESC

Apply for job page to select job and business details.

Fields/Calculations	*
Tables/Queries	job, business
Search Criteria	job.jobID = '\$jobID' and job.businessID = business.businessID
Grouping	
Having	
Sort Order	

Apply for job page to check application does not already exist.

Fields/Calculations	*
Tables/Queries	application
Search Criteria	userID = '\$userID' AND jobID = '\$jobID'
Grouping	
Having	
Sort Order	

Apply for job page to create a new application. *(Insert query)*

Fields/Calculations	userID, jobID, coverLetter
Tables/Queries	application
Search Criteria	
Grouping	
Having	
Sort Order	

Applications page to get all the user's existing applications.

Fields/Calculations	*
Tables/Queries	application, job, business
Search Criteria	userID = '\$userID' AND application.jobID = job.jobID AND job.businessID = business.businessID
Grouping	
Having	
Sort Order	applicationID DESC

Business home page to display 3 most recent job listings.

Fields/Calculations	*
Tables/Queries	Job
Search Criteria	businessID = '\$businessID'
Grouping	
Having	
Sort Order	jobID DESC
Limit	3

View listings page to display all listings for the business.

Fields/Calculations	*
Tables/Queries	job
Search Criteria	businessID = '\$businessID'
Grouping	
Having	
Sort Order	jobID DESC

View applicants page to get job details.

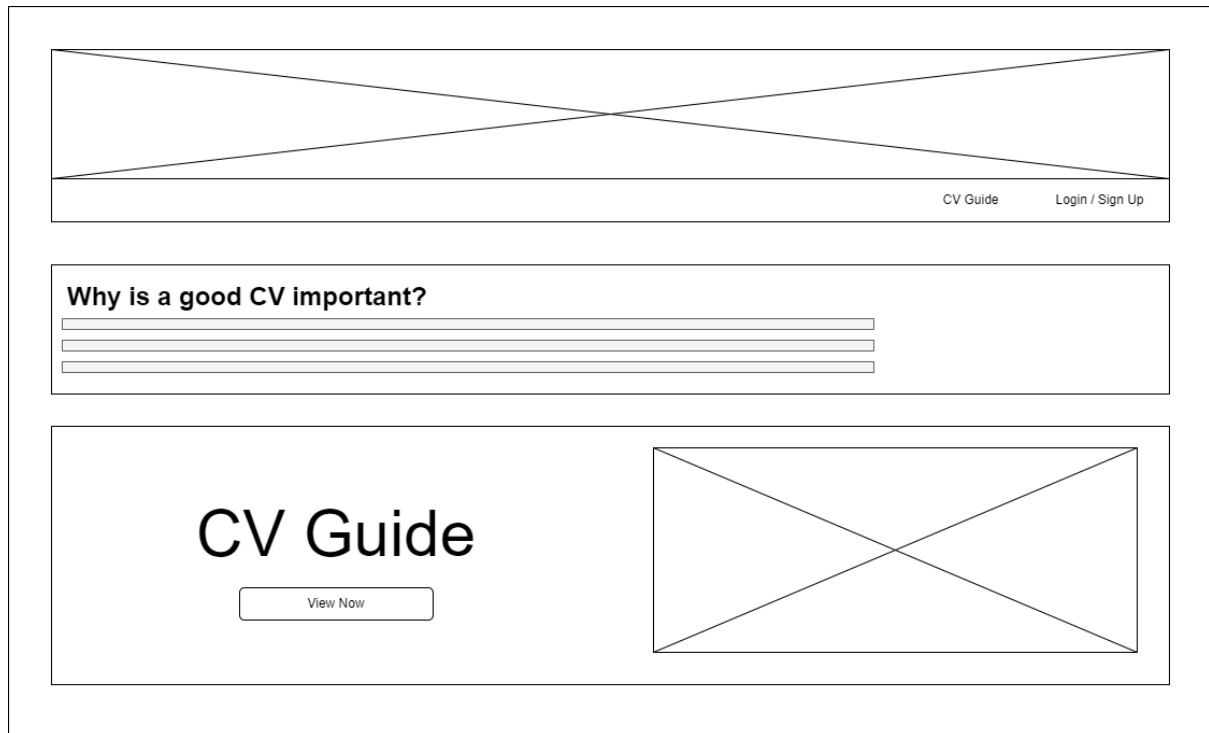
Fields/Calculations	*
Tables/Queries	job
Search Criteria	job.jobID = '\$jobID'
Grouping	
Having	
Sort Order	

View applicants page to get details for each applicant.

Fields/Calculations	*
Tables/Queries	user, application
Search Criteria	application.jobID = '\$jobID' AND user.userID = application.userID
Grouping	
Having	
Sort Order	

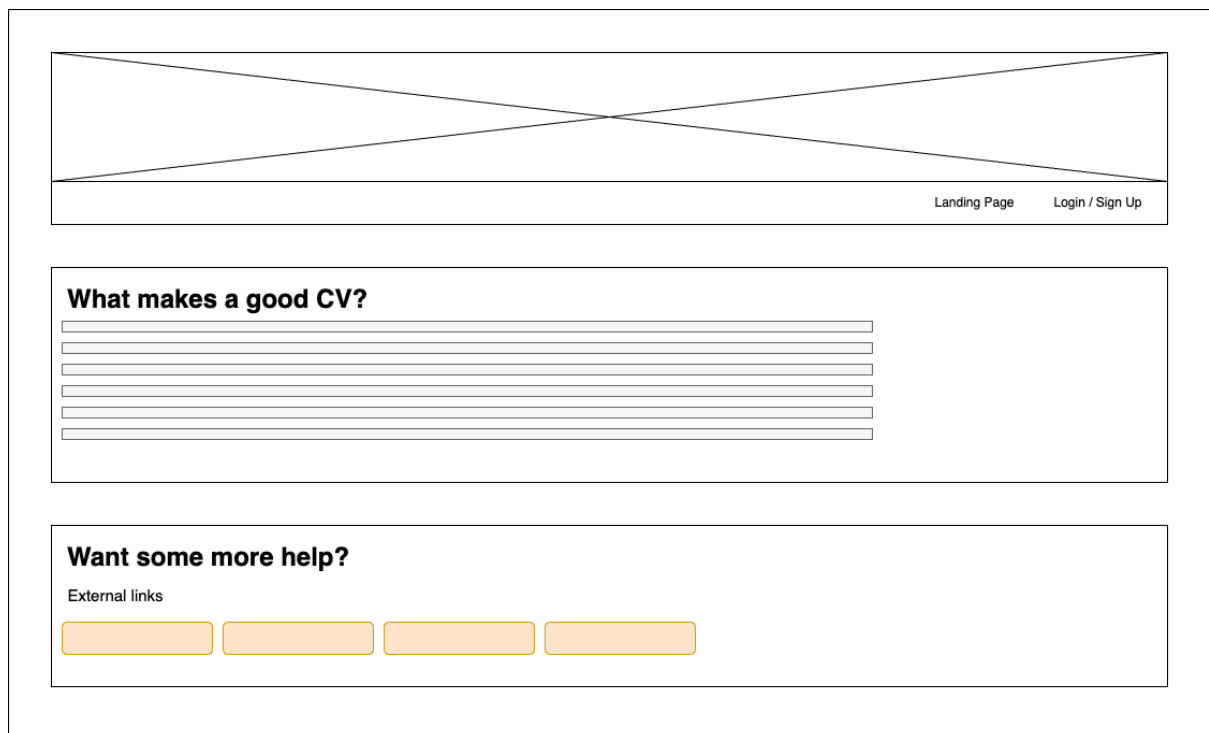
2.6 Wireframes

2.6.1 Landing Page

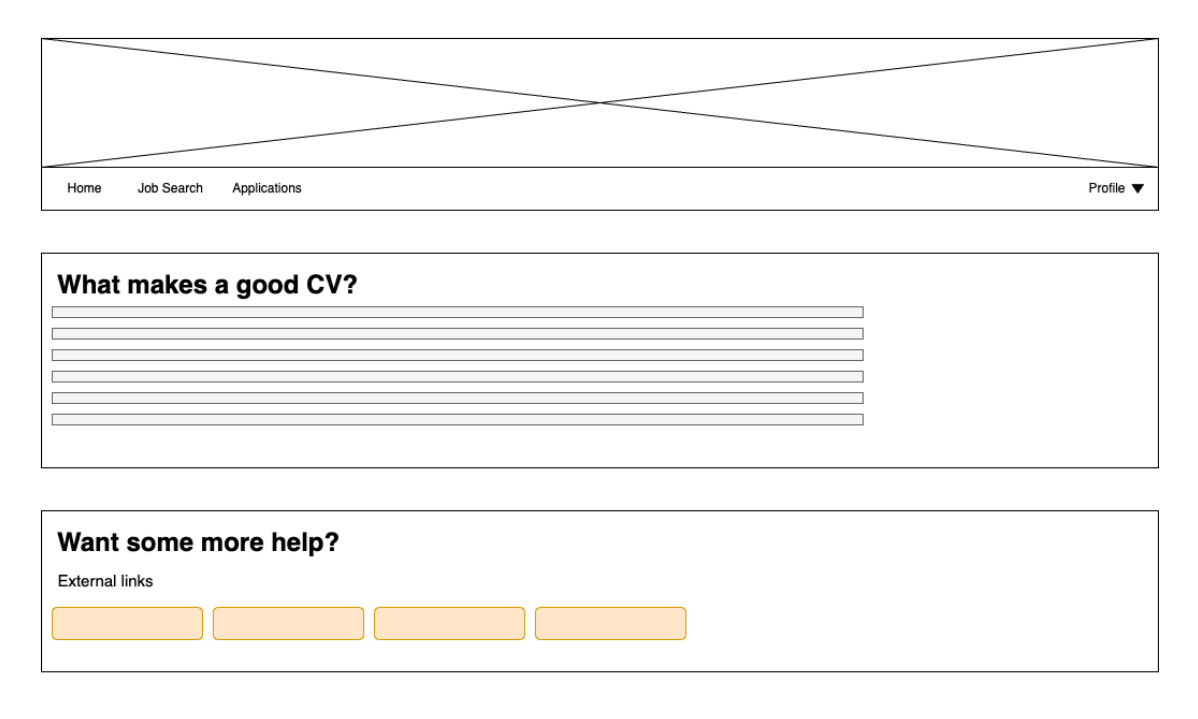


2.6.2 CV Guide Page

2.6.2.a When the user has not already logged in.



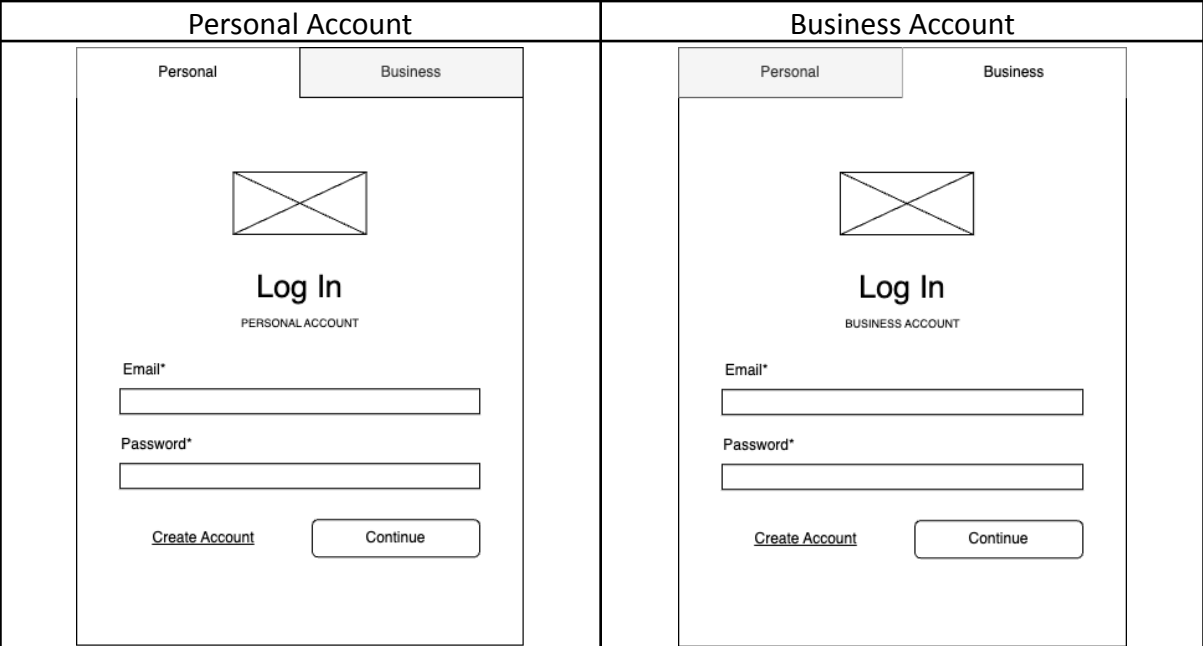
2.6.2.b When the user has logged in.



The layout shows a user profile page after login. At the top is a navigation bar with links for 'Home', 'Job Search', and 'Applications', and a 'Profile' dropdown menu. Below this is a section titled 'What makes a good CV?' followed by seven horizontal lines for text input. The next section is titled 'Want some more help?' and includes the text 'External links' above four orange rectangular buttons.

2.6.3 Login Page

The login pages will have a tab bar along the top which allows the user to select their account type. The tab bar will be colour coded to indicate which type of account is currently selected. The login pages will contain links to the sign-up pages. This link should stand out less than the 'Continue' button.



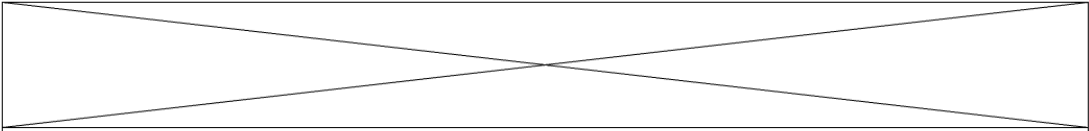
This block contains two side-by-side wireframes for login pages. The left wireframe is for a 'Personal Account' and the right is for a 'Business Account'. Both pages feature a tab bar at the top with 'Personal' and 'Business' tabs. The 'Business' tab is highlighted in grey. Below the tabs is a placeholder for a logo (a rectangle with an 'X'). The main heading is 'Log In', followed by the account type ('PERSONAL ACCOUNT' or 'BUSINESS ACCOUNT'). Below this are input fields for 'Email*' and 'Password*'. At the bottom, there is a 'Create Account' link and a 'Continue' button.

2.6.4 Sign Up Page

The sign-up pages will have a tab bar along the top which allows the user to select a personal account or a business account. The tab bar will be colour coded to indicate which type of account is currently selected.

Personal Account	Business Account
PersonalBusiness  Sign Up PERSONAL ACCOUNT Name* Email* Password* Continue	PersonalBusiness  Sign Up BUSINESS ACCOUNT Business Name* Email* Phone Number* Password* Continue

2.6.5 User Home Page



HomeJob SearchApplications

Profile ▼

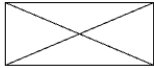
Recent Job Postings

Apply Now

Apply Now

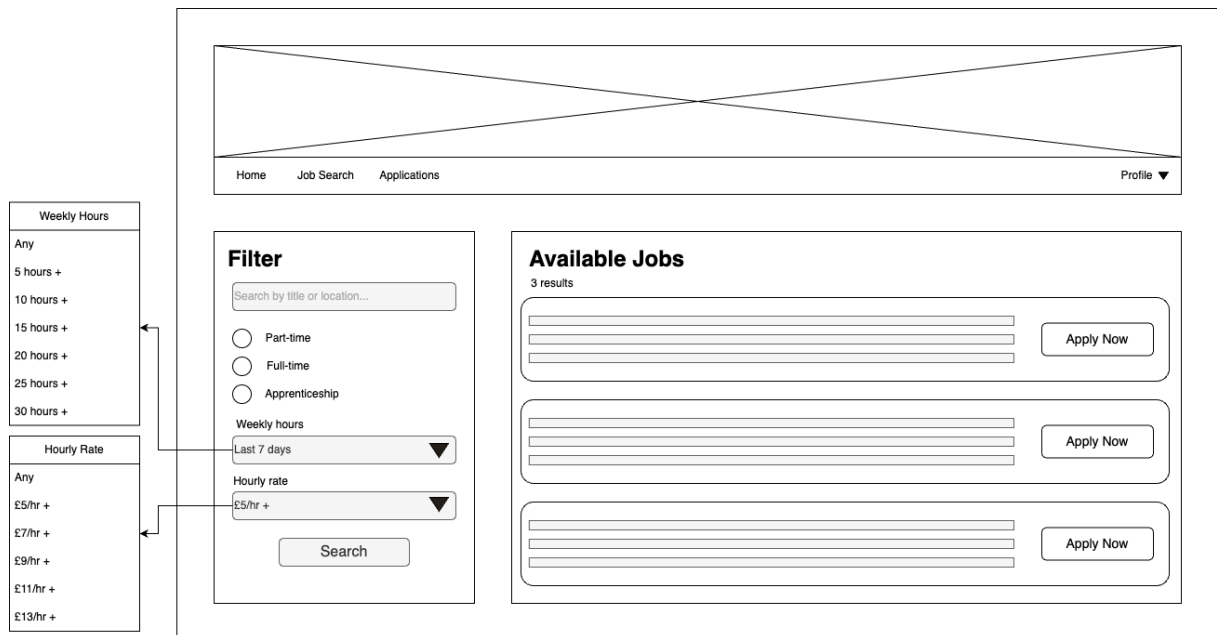
Apply Now

CV Guide



CV Guide

2.6.6 Job Search Page



The Job Search Page mockup features a navigation bar with links to Home, Job Search, and Applications, and a Profile dropdown. On the left, there are two filter panels: 'Weekly Hours' and 'Hourly Rate'. The 'Weekly Hours' panel lists options from 'Any' to '30 hours +'. The 'Hourly Rate' panel lists options from 'Any' to '£13/hr +'. The main 'Filter' section includes a search input, radio buttons for 'Part-time', 'Full-time', and 'Apprenticeship', and dropdowns for 'Weekly hours' (set to 'Last 7 days') and 'Hourly rate' (set to '£5/hr +'). A 'Search' button is at the bottom of the filter section. The 'Available Jobs' section shows '3 results' and lists three job entries, each with an 'Apply Now' button. Arrows indicate that the filter panels are linked to the corresponding dropdowns in the main filter section.

Weekly Hours

Any

5 hours +

10 hours +

15 hours +

20 hours +

25 hours +

30 hours +

Hourly Rate

Any

£5/hr +

£7/hr +

£9/hr +

£11/hr +

£13/hr +

Filter

Search by title or location...

☐ Part-time

☐ Full-time

☐ Apprenticeship

Weekly hours

Last 7 days

Hourly rate

£5/hr +

Search

Available Jobs

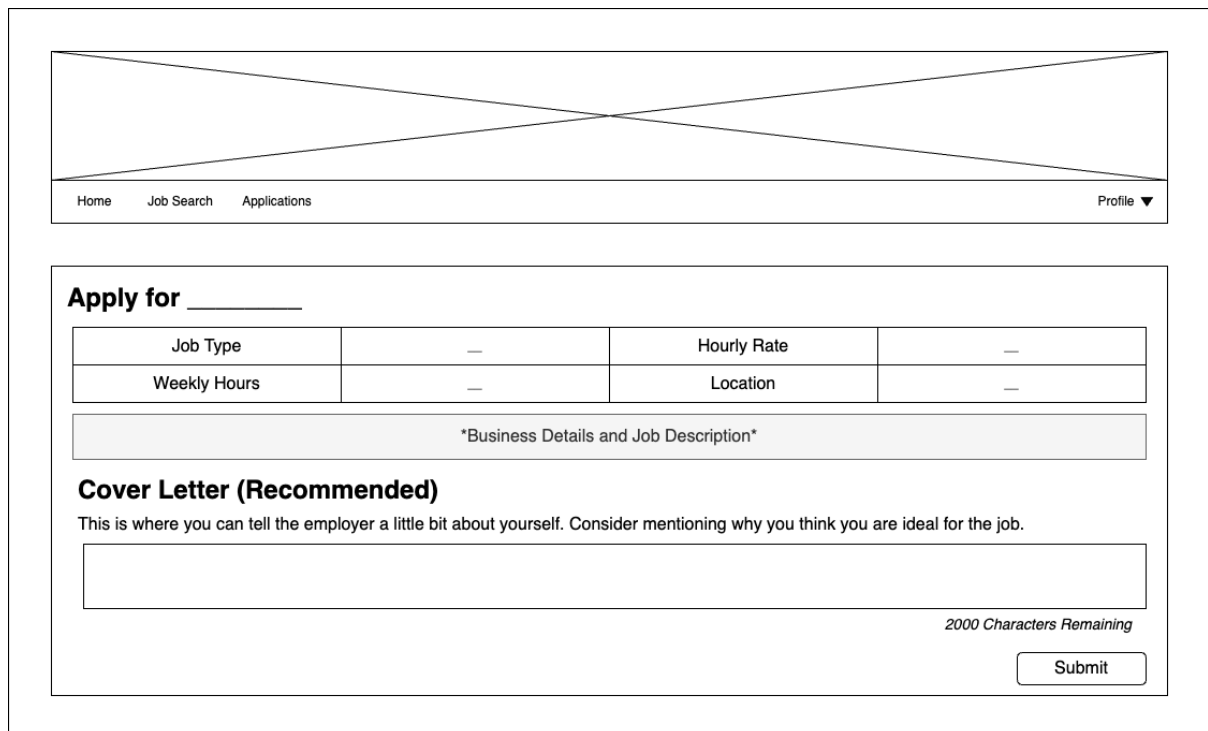
3 results

Apply Now

Apply Now

Apply Now

2.6.7 Apply for Job Page



The Apply for Job Page mockup features a navigation bar with links to Home, Job Search, and Applications, and a Profile dropdown. The main section is titled 'Apply for _____'. It contains a table with four columns: Job Type, Hourly Rate, Weekly Hours, and Location. Below the table is a section for 'Business Details and Job Description'. The 'Cover Letter (Recommended)' section includes a text area for the user to write a cover letter, with a character count of '2000 Characters Remaining' and a 'Submit' button.

Home Job Search Applications Profile ▼

Apply for _____

Job Type	Hourly Rate	Weekly Hours	Location
—	—	—	—

Business Details and Job Description

Cover Letter (Recommended)

This is where you can tell the employer a little bit about yourself. Consider mentioning why you think you are ideal for the job.

2000 Characters Remaining

Submit

2.6.8 View Applications Page

HomeJob SearchApplicationsProfile ▼

Your Applications

Job Title

Job description and cover letter

Job Title

Job description and cover letter

Job Title

Job description and cover letter

2.6.9 Profile Page

2.6.9.a Personal Account

HomeJob SearchApplicationsProfile ▼

Your Profile

PERSONAL ACCOUNT

Name

Email

Log Out

2.6.9.b Business Account

HomeCreate JobView ListingsProfile ▼

Your Profile

BUSINESS ACCOUNT

Name

Email

Phone Number

Log Out

2.6.10 Business Home Page

HomeCreate JobView ListingsProfile ▼

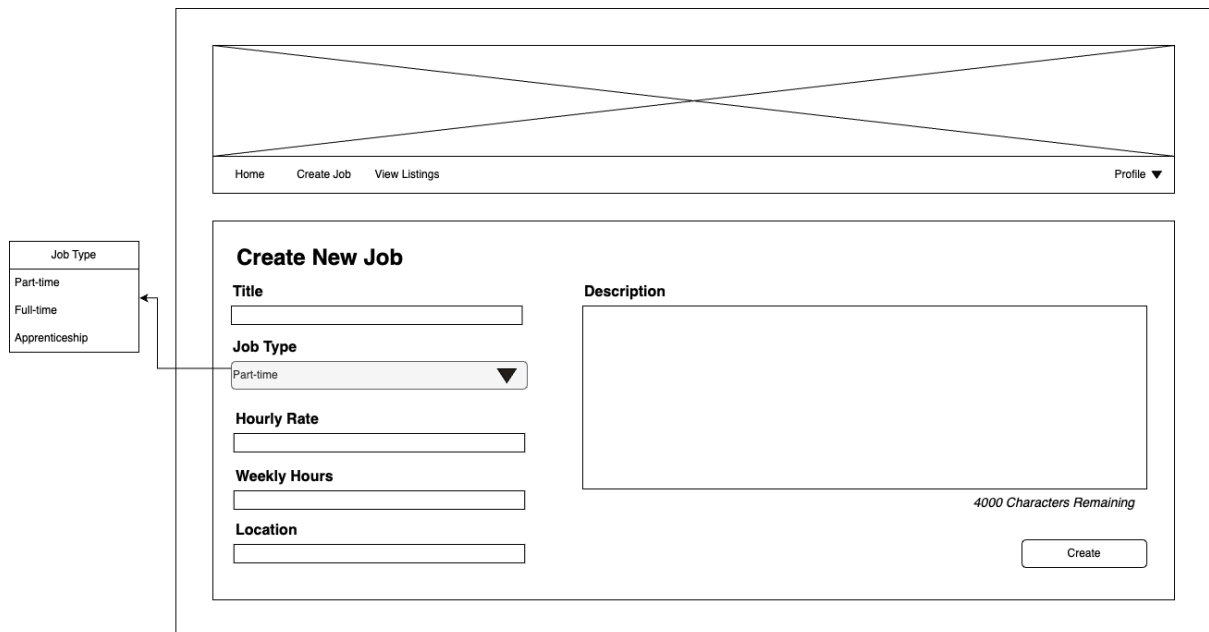
Your Recent Listings

View Applicants

View Applicants

View Applicants

2.6.11 Create New Job Page



The 'Create New Job' page features a navigation bar with links for Home, Create Job, View Listings, and a Profile dropdown. A 'Job Type' sidebar on the left lists Part-time, Full-time, and Apprenticeship, with an arrow pointing to the 'Job Type' dropdown on the main form. The form includes fields for Title, Job Type (set to Part-time), Hourly Rate, Weekly Hours, and Location. A large text area for Description has a '4000 Characters Remaining' indicator. A 'Create' button is at the bottom right.

Job Type
Part-time
Full-time
Apprenticeship

Create New Job

Title

Job Type
Part-time ▼

Hourly Rate

Weekly Hours

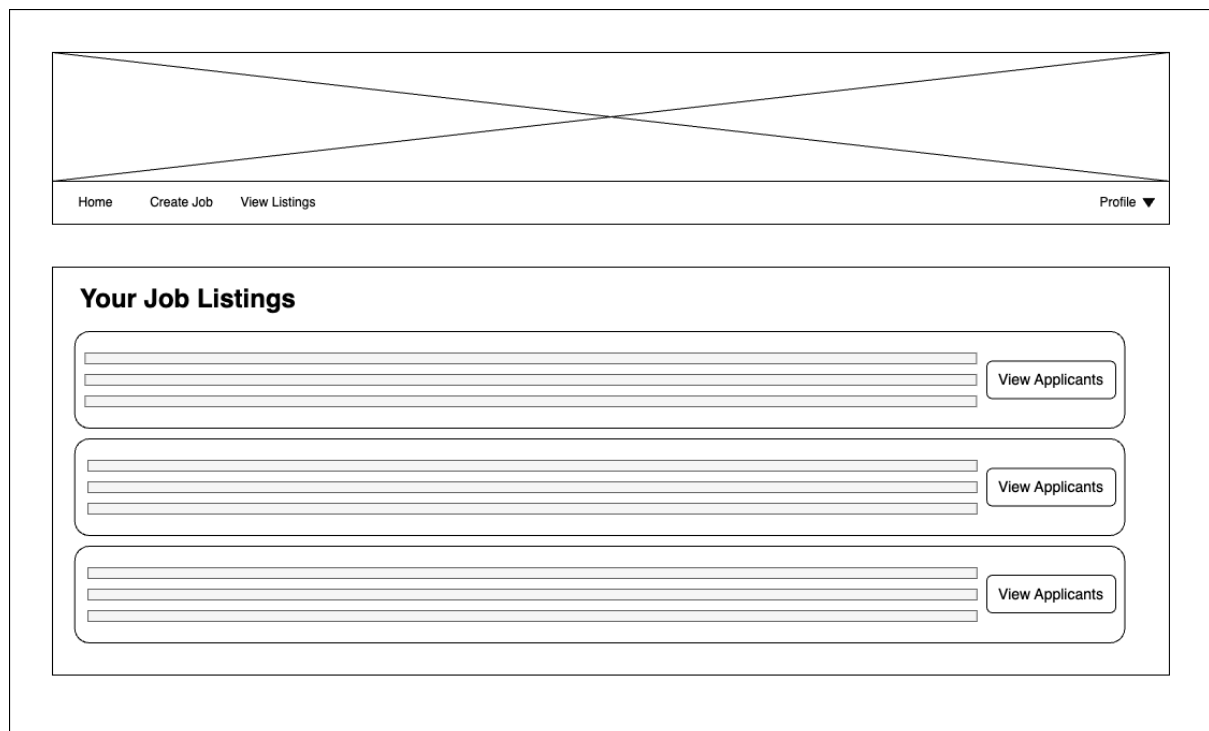
Location

Description

4000 Characters Remaining

Create

2.6.12 View Job Listings Page



The 'View Job Listings' page features a navigation bar with links for Home, Create Job, View Listings, and a Profile dropdown. The main content area, titled 'Your Job Listings', displays three job listing cards. Each card contains placeholder text for job details and a 'View Applicants' button.

Your Job Listings

View Applicants

View Applicants

View Applicants

2.6.13 View Applicants Page

The wireframe illustrates the layout of the 'View Applicants' page. At the top, a navigation bar contains links for 'Home', 'Create Job', and 'View Listings', with a 'Profile' dropdown menu on the right. Below the navigation bar, the page is divided into two main sections. The left section, titled 'Job Title', contains a large text area labeled '*Job Description*'. The right section, titled 'Applicants', contains two rounded rectangular boxes, each labeled '*User details and cover letter*'. To the left of the 'Applicants' section, there is a 'Job Details' section with a list of horizontal lines representing details.

2.7 Pseudocode Design

2.7.1 CV Guide Page

1. START SESSION
2. IF 'accountID' SESSION VARIABLE IS NOT SET
3. DISPLAY NAV BAR FOR GUEST USERS (NOT LOGGED IN)
4. ELSE
5. DISPLAY NAV BAR FOR LOGGED IN USERS
6. END IF

2.7.2 Login Page

User Login Page

1. START SESSION
2. SET ERROR REPORTING TO FATAL AND PARSE ONLY
3. IF 'newAccount' SESSION VARIABLE IS SET
4. DISPLAY NEW ACCOUNT MESSAGE
5. DELETE 'newAccount' SESSION VARIABLE
6. END IF
7. IF 'accountID' SESSION VARIABLE IS SET
8. IF 'accountType' SESSION VARIABLE IS 'Personal'
9. NAVIGATE TO 'userHome.php'
10. ELSE IF 'accountType' SESSION VARIABLE IS 'Business'
11. NAVIGATE TO 'businessHome.php'
12. END IF
13. END IF
14. IF 'loginBtn' POST VARIABLE IS SET
15. ESTABLISH DATABASE CONNECTION VARIABLES
16. ATTEMPT TO CONNECT TO THE DATABASE
17. IF CONNECTION ERROR
18. DISPLAY CONNECTION ERROR DETAILS
19. END PHP SCRIPT
20. END IF
21. CREATE VARIABLES USING POST FROM HTML FORM
22. CREATE QUERY TO SEARCH DATABASE USING THE ACCOUNT DETAILS ENTERED
23. EXECUTE THE QUERY
24. IF THE QUERY IS SUCCESSFUL
25. COUNT NUMBER OF ROWS RETURNED
26. IF 1 ROW IS RETURNED
27. STORE ROW IN ASSOCIATIVE ARRAY
28. SET SESSION VARIABLES 'accountID' and 'accountType'
29. NAVIGATE TO 'userHome.php'
30. ELSE
31. DISPLAY ACCOUNT NOT FOUND MESSAGE
32. END IF
33. ELSE
34. DISPAY QUERY ERROR MESSAGE
35. END IF
36. CLOSE DATABASE CONNECTION
37. END IF

Business Login Page

1. START SESSION
2. SET ERROR REPORTING TO FATAL AND PARSE ONLY
3. IF 'newAccount' SESSION VARIABLE IS SET
4. DISPLAY NEW ACCOUNT MESSAGE
5. DELETE 'newAccount' SESSION VARIABLE
6. END IF
7. IF 'accountID' SESSION VARIABLE IS SET
8. IF 'accountType' SESSION VARIABLE IS 'Personal'
9. NAVIGATE TO 'userHome.php'
10. ELSE IF 'accountType' SESSION VARIABLE IS 'Business'
11. NAVIGATE TO 'businessHome.php'
12. END IF
13. END IF
14. IF 'logInBtn' POST VARIABLE IS SET
15. ESTABLISH DATABASE CONNECTION VARIABLES
16. ATTEMPT TO CONNECT TO THE DATABASE
17. IF CONNECTION ERROR
18. DISPLAY CONNECTION ERROR DETAILS
19. END PHP SCRIPT
20. END IF
21. CREATE VARIABLES USING POST FROM HTML FORM
22. CREATE QUERY TO SEARCH DATABASE USING THE ACCOUNT DETAILS ENTERED
23. EXECUTE THE QUERY
24. IF THE QUERY IS SUCCESSFUL
25. COUNT NUMBER OF ROWS RETURNED
26. IF 1 ROW IS RETURNED
27. STORE ROW IN ASSOCIATIVE ARRAY
28. SET SESSION VARIABLES 'accountID' and 'accountType'
29. NAVIGATE TO 'businessHome.php'
30. ELSE
31. DISPLAY ACCOUNT NOT FOUND MESSAGE
32. END IF
33. ELSE
34. DISPAY QUERY ERROR MESSAGE
35. END IF
36. CLOSE DATABASE CONNECTION
37. END IF

2.7.3 Sign Up Page

User Sign Up Page

1. START SESSION
2. IF 'accountID' SESSION VARIABLE IS SET
3. IF 'accountType' SESSION VARIABLE IS 'Personal'
4. NAVIGATE TO 'userHome.php'
5. ELSE IF 'accountType' SESSION VARIABLE IS 'Business'
6. NAVIGATE TO 'businessHome.php'
7. END IF
8. END IF
9. SET ERROR REPORTING TO FATAL AND PARSE ONLY
10. IF 'signUpBtn' POST VARIABLE IS SET
11. ESTABLISH DATABASE CONNECTION VARIABLES
12. ATTEMPT TO CONNECT TO THE DATABASE
13. IF CONNECTION ERROR
14. DISPLAY CONNECTION ERROR DETAILS
15. END PHP SCRIPT
16. END IF
17. CREATE VARIABLES USING POST FROM HTML FORM
18. CREATE QUERY TO CHECK USER DOES NOT ALREADY EXIST
19. EXECUTE THE QUERY
20. IF THE QUERY IS SUCCESSFUL
21. COUNT NUMBER OF ROWS RETURNED
22. IF 0 ROWS ARE RETURNED
23. CREATE QUERY TO INSERT NEW USER TO DATABASE
24. EXECUTE THE QUERY
25. IF THE QUERY IS SUCCESSFUL
26. SET 'newAccount' SESSION VARIABLE TO 'True'
27. NAVIGATE TO 'userLogin.php'
28. ELSE
29. DISPLAY ERROR CREATING ACCOUNT MESSAGE
30. END IF
31. ELSE
32. DISPLAY ACCOUNT ALREADY EXISTS MESSAGE
33. END IF
34. ELSE
35. DISPLAY QUERY ERROR MESSAGE
36. END IF
37. CLOSE DATABASE CONNECTION
38. END IF

Business Sign Up Page

1. START SESSION
2. IF 'accountID' SESSION VARIABLE IS SET
3. IF 'accountType' SESSION VARIABLE IS 'Personal'
4. NAVIGATE TO 'userHome.php'
5. ELSE IF 'accountType' SESSION VARIABLE IS 'Business'
6. NAVIGATE TO 'businessHome.php'
7. END IF
8. END IF
9. SET ERROR REPORTING TO FATAL AND PARSE ONLY

10. IF 'signUpBtn' POST VARIABLE IS SET
11. ESTABLISH DATABASE CONNECTION VARIABLES
12. ATTEMPT TO CONNECT TO THE DATABASE
13. IF CONNECTION ERROR
14. DISPLAY CONNECTION ERROR DETAILS
15. END PHP SCRIPT
16. END IF
17. CREATE VARIABLES USING POST FROM HTML FORM
18. CREATE QUERY TO CHECK BUSINESS DOES NOT ALREADY EXIST
19. EXECUTE THE QUERY
20. IF THE QUERY IS SUCCESSFUL
21. COUNT NUMBER OF ROWS RETURNED
22. IF 0 ROWS ARE RETURNED

23. CREATE QUERY TO INSERT NEW BUSINESS TO DATABASE
24. EXECUTE THE QUERY
25. IF THE QUERY IS SUCCESSFUL
26. SET 'newAccount' SESSION VARIABLE TO 'True'
27. NAVIGATE TO 'businessLogin.php'
28. ELSE
29. DISPLAY ERROR CREATING ACCOUNT MESSAGE
30. END IF

31. ELSE
32. DISPLAY ACCOUNT ALREADY EXISTS MESSAGE
33. END IF
34. ELSE
35. DISPAY QUERY ERROR MESSAGE
36. END IF
37. CLOSE DATABASE CONNECTION
38. END IF

2.7.4 User Home Page

1. START SESSION
2. IF 'logOutBtn' POST VARIABLE IS SET
3. DESTORY SESSION
4. NAVIGATE TO 'index.html'
5. END IF

6. IF 'accountID' SESSION VARIABLE IS NOT SET
7. NAVIGATE TO 'userLogin.php'
8. END IF
9. IF 'accountType' SESSION VARIABLE IS 'Business'
10. NAVIGATE TO 'businessHome.php'
11. END IF

12. SET ERROR REPORTING TO FATAL AND PARSE ONLY
13. ESTABLISH DATABASE CONNECTION VARIABLES
14. ATTEMPT TO CONNECT TO THE DATABASE
15. IF CONNECTION ERROR
16. DISPLAY CONNECTION ERROR DETAILS
17. END PHP SCRIPT
18. END IF
19. CREATE QUERY TO GET 3 MOST RECENT JOBS
20. EXECUTE THE QUERY
21. IF THE QUERY IS SUCCESSFUL
22. COUNT NUMBER OF ROWS RETURNED
23. IF MORE THAN 0 ROWS ARE RETURNED
24. START LOOP FOR EACH ROW RETURNED
25. SET VARIABLES FROM DATABASE
26. DISPLAY CARD WITH JOB DETAILS AND APPLY NOW BUTTON
27. END LOOP
28. ELSE
29. DISPLAY MESSAGE SAYING NO JOBS WERE FOUND
30. END IF
31. ELSE
32. DISPLAY QUERY ERROR MESSAGE
33. END IF
34. CLOSE DATABASE CONNECTION

2.7.5 Job Search Page

1. START SESSION
2. IF 'logOutBtn' POST VARIABLE IS SET
3. DESTORY SESSION
4. NAVIGATE TO 'index.html'
5. END IF

6. IF 'accountID' SESSION VARIABLE IS NOT SET
7. NAVIGATE TO 'userLogin.php'
8. END IF
9. IF 'accountType' SESSION VARIABLE IS 'Business'
10. NAVIGATE TO 'businessHome.php'
11. END IF

12. IF 'searchButton' POST VARIABLE IS SET
13. SET ERROR REPORTING TO FATAL AND PARSE ONLY
35. ESTABLISH DATABASE CONNECTION VARIABLES
36. ATTEMPT TO CONNECT TO THE DATABASE
37. IF CONNECTION ERROR
38. DISPLAY CONNECTION ERROR DETAILS
39. END PHP SCRIPT
40. END IF
14. CREATE VARIABLES WITH DATA FROM HTML FORM
15. CREATE QUERY TO GET JOBS MATCHING SEARCH OPTIONS
16. EXECUTE THE QUERY

17. IF THE QUERY IS SUCCESSFUL
18. COUNT NUMBER OF ROWS RETURNED
19. DISPLAY NUMBER OF ROWS RETURNED
20. IF MORE THAN 0 ROWS ARE RETURNED
21. START LOOP FOR EACH ROW RETURNED
22. CREATE VARIABLES WITH DATA FROM DATABASE
23. DISPLAY CARD WITH JOB DETAILS AND APPLY NOW BUTTON
24. END LOOP
25. ELSE
26. DISPLAY MESSAGE SAYING NO JOBS WERE FOUND
27. END IF
28. ELSE
29. DISPLAY QUERY ERROR MESSAGE
30. END IF

31. CLOSE DATABASE CONNECTION
32. ELSE
33. DISPLAY MESSAGE SAYING TO MAKE A SEARCH TO FIND JOBS
34. END IF

2.7.6 Apply for Job Page

1. START SESSION
2. IF 'logOutBtn' POST VARIABLE IS SET
3. DESTORY SESSION
4. NAVIGATE TO 'index.html'
5. END IF

6. IF 'accountID' SESSION VARIABLE IS NOT SET
7. NAVIGATE TO 'userLogin.php'
8. END IF
9. IF 'accountType' SESSION VARIABLE IS 'Business'
10. NAVIGATE TO 'businessHome.php'
11. END IF

12. SET ERROR REPORTING TO FATAL AND PARSE ONLY
13. ESTABLISH DATABASE CONNECTION VARIABLES
14. ATTEMPT TO CONNECT TO THE DATABASE
15. IF CONNECTION ERROR
16. DISPLAY CONNECTION ERROR DETAILS
17. END PHP SCRIPT
18. END IF
19. GET 'jobID' USING GET METHOD
20. CREATE QUERY TO GET DETAILS FOR SELECTED JOB
21. EXECUTE THE QUERY

22. IF THE QUERY IS SUCCESSFUL
23. COUNT THE NUMBER OF ROWS RETURNED
24. IF MORE THAN 0 ROWS RETURNED
25. STORE JOB DETAILS IN AN ASSOCIATIVE ARRAY
26. CREATE VARIABLES USING DATA FROM DATABASE
27. IF 'submitCover' POST VARIABLE IS SET
28. CREATE VARIABLES FOR 'userID' and 'coverLetter'
29. CREATE QUERY TO CHECK APPLICATION HAS NOT ALREADY BEEN MADE
30. EXECUTE THE QUERY

31. IF THE QUERY IS SUCCESSFUL
32. COUNT THE NUMBER OF ROWS RETURNED
33. IF 0 ROWS ARE RETURNED
34. CREATE QUERY TO INSERT APPLICATION TO DATABASE
35. EXECUTE THE QUERY
36. IF THE QUERY IS SUCCESSFUL
37. DISPLAY APPLICATION SUCCESS MESSAGE
38. ELSE
39. DISPLAY QUERY ERROR MESSAGE
40. END IF
41. ELSE

```

42.         DISPLAY MESSAGE SAYING THE USER HAS ALREADY APPLIED FOR JOB
43.     END IF
44. ELSE
45.     DISPLAY QUERY ERROR MESSAGE
46. END IF
47. END IF

48.     DISPLAY CARD WITH JOB DETAILS AND DESCRIPTION
49. ELSE
50.     DISPLAY MESSAGE SAYING JOB WAS NOT FOUND IN DATABASE
51. END IF
52. ELSE
53.     DISPLAY QUERY ERROR MESSAGE
54. END IF
55. CLOSE DATABASE CONNECTION

```

2.7.7 View Applications Page

```

1.  START SESSION
2.  IF 'logOutBtn' POST VARIABLE IS SET
3.      DESTORY SESSION
4.      NAVIGATE TO 'index.html'
5.  END IF

6.  IF 'accountID' SESSION VARIABLE IS NOT SET
7.      NAVIGATE TO 'userLogin.php'
8.  END IF
9.  IF 'accountType' SESSION VARIABLE IS 'Business'
10.     NAVIGATE TO 'businessHome.php'
11. END IF

12. SET ERROR REPORTING TO FATAL AND PARSE ONLY
13. ESTABLISH DATABASE CONNECTION VARIABLES
14. ATTEMPT TO CONNECT TO THE DATABASE
15. IF CONNECTION ERROR
16.     DISPLAY CONNECTION ERROR DETAILS
17. END PHP SCRIPT
18. END IF
19. CREATE 'userID' VARIABLE
20. CREATE QUERY TO SELECT ALL EXISTING APPLICATIONS FOR CURRENT USER
21. EXECUTE THE QUERY
22. IF THE QUERY IS SUCCESSFUL
23.     COUNT THE NUMBER OF ROWS RETURNED
24.     IF MORE THAN 0 ROWS ARE RETURNED
25.         START LOOP FOR EACH ROW RETURNED
26.             CREATE VARIABLES USING DATA FROM DATABASE
27.         END LOOP

```

28. IF 'coverLetter' IS BLANK
29. CREATE MESSAGE TO SAY NO COVER LETTER WAS PROVIDED
30. END IF
31. DISPLAY CARD WITH JOB DETAILS AND COVER LETTER
32. ELSE
33. DISPLAY MESSAGE SAYING USER HAS NOT APPLIED TO ANY JOBS YET
34. END IF
35. ELSE
36. DISPLAY QUERY ERROR MESSAGE
37. CLOSE DATABASE CONNECTION

2.7.8 Profile Page

1. START SESSION
2. IF 'accountID' SESSION VARIABLE IS SET
3. SET ERROR REPORTING TO FATAL AND PARSE ONLY
4. IF 'logOutBtn' POST VARIABLE IS SET
5. DESTROY SESSION
6. NAVIGATE TO 'index.html'
7. ELSE
8. ESTABLISH DATABASE CONNECTION VARIABLES
9. ATTEMPT TO CONNECT TO THE DATABASE
10. IF CONNECTION ERROR
11. DISPLAY CONNECTION ERROR DETAILS
12. END PHP SCRIPT
13. END IF
14. SET 'accountID' VARIABLE FROM SESSION
15. END IF
16. END IF

17. IF 'accountType' SESSION VARIABLE IS 'Personal'
18. DISPLAY USER BANNER IMAGE
19. DISPLAY USER NAV BAR
20. ELSE IF 'accountType' SESSION VARIABLE IS 'Business'
21. DISPLAY BUSINESS BANNER IMAGE
22. DISPLAY BUSINESS NAV BAR
23. END IF

24. IF 'accountType' SESSION VARIABLE IS 'Personal'
25. CREATE QUERY TO GET USER DETAILS FROM USER TABLE IN DATABASE
26. EXECUTE THE QUERY
27. IF THE QUERY IS SUCCESSFUL
28. COUNDTHE NUMBER OF ROWS RETURNED
29. IF 1 ROW IS RETURNED
30. STORE DETAILS FROM DATABASE IN ASSOCIATE ARRAY
31. DISPLAY USER ACCOUNT DETAILS
32. ELSE

```

33.     DISPLAY ERROR MESSAGE SAYING USER COULD NOT BE FOUND
34.     END IF
35. ELSE
36.     DISPLAY QUERY ERROR MESSAGE
37. END IF
38. ELSE IF 'accountType' SESSION VARIABLE IS 'Business'
39.     CREATE QUERY TO GET BUSINESS DETAILS FROM BUSINESS TABLE IN DATABASE
40.     EXECUTE THE QUERY
41.     IF THE QUERY IS SUCCESSFUL
42.         COUNT THE NUMBER OF ROWS RETURNED
43.         IF 1 ROW IS RETURNED
44.             STORE DETAILS FROM DATABASE IN ASSOCIATE ARRAY
45.             DISPLAY BUSINESS ACCOUNT DETAILS
46.         ELSE
47.             DISPLAY ERROR MESSAGE SAYING BUSINESS COULD NOT BE FOUND
48.         END IF
49.     ELSE
50.         DISPLAY QUERY ERROR MESSAGE
51.     END IF
52. END IF
53. CLOSE DATABASE CONNECTION

```

2.7.9 Business Home Page

```

1.  START SESSION
2.  IF 'logOutBtn' POST VARIABLE IS SET
3.      DESTORY SESSION
4.      NAVIGATE TO 'index.html'
5.  END IF

6.  IF 'accountID' SESSION VARIABLE IS NOT SET
7.      NAVIGATE TO 'userLogin.php'
8.  END IF
9.  IF 'accountType' SESSION VARIABLE IS 'Personal'
10.     NAVIGATE TO 'userHome.php'
11. END IF

12. SET ERROR REPORTING TO FATAL AND PARSE ONLY
13. ESTABLISH DATABASE CONNECTION VARIABLES
14. ATTEMPT TO CONNECT TO THE DATABASE
15. IF CONNECTION ERROR
16.     DISPLAY CONNECTION ERROR DETAILS
17. END PHP SCRIPT
18. END IF
19. CREATE 'businessID' VARIABLE FROM SESSION
20. CREATE QUERY TO GET BUSINESS' 3 MOST RECENT JOB POSTINGS
21. EXECUTE THE QUERY

```

22. IF CONNECTION ERROR
23. DISPLAY CONNECTION ERROR DETAILS
24. END PHP SCRIPT
25. END IF
26. IF THE QUERY IS SUCCESSFUL
27. COUNT THE NUMBER OF ROWS RETURNED
28. IF MORE THAN 0 ROWS ARE RETURNED
29. START LOOP FOR EACH ROW RETURNED
30. CREATE VARIABLES USING DATA FROM DATABASE
31. DISPLAY JOB DETAILS AND VIEW APPLICANTS BUTTON IN JOB CARD
32. END LOOP
33. ELSE
34. DISPLAY MESSAGE SAYING BUSINESS HAS NOT POSTED ANY JOBS YET
35. END IF
36. ELSE
37. DISPLAY QUERY ERROR MESSAGE
38. END IF
39. CLOSE DATABASE CONNECTION

2.7.10 Create New Job Page

1. START SESSION
2. IF 'logOutBtn' POST VARIABLE IS SET
3. DESTORY SESSION
4. NAVIGATE TO 'index.html'
5. END IF

6. IF 'accountID' SESSION VARIABLE IS NOT SET
7. NAVIGATE TO 'userLogin.php'
8. END IF
9. IF 'accountType' SESSION VARIABLE IS 'Personal'
10. NAVIGATE TO 'userHome.php'
11. END IF

12. IF 'createJob' POST VARIABLE IS SET
13. SET ERROR REPORTING TO FATAL AND PARSE ONLY
40. ESTABLISH DATABASE CONNECTION VARIABLES
41. ATTEMPT TO CONNECT TO THE DATABASE
42. IF CONNECTION ERROR
43. DISPLAY CONNECTION ERROR DETAILS
44. END PHP SCRIPT
45. END IF
46. CREATE 'businessID' VARIABLE FROM SESSION
14. CREATE VARIABLES USING DATA FROM HTML FORM
15. CREATE QIERY TO INSERT JOB INTO DATABASE
16. EXECUTE THE QUERY
17. IF THE QUERY IS SUCCESSFUL

18. DISPLAY SUCCESS MESSAGE
19. ELSE
20. DISPLAY QUERY ERROR MESSAGE
21. END IF
22. CLOSE DATABASE CONNECTION
23. END IF

2.7.11 View Job Listings Page

1. START SESSION
2. IF 'logOutBtn' POST VARIABLE IS SET
3. DESTORY SESSION
4. NAVIGATE TO 'index.html'
5. END IF

6. IF 'accountID' SESSION VARIABLE IS NOT SET
7. NAVIGATE TO 'userLogin.php'
8. END IF
9. IF 'accountType' SESSION VARIABLE IS 'Personal'
10. NAVIGATE TO 'userHome.php'
11. END IF

12. SET ERROR REPORTING TO FATAL AND PARSE ONLY
13. ESTABLISH DATABASE CONNECTION VARIABLES
14. ATTEMPT TO CONNECT TO THE DATABASE
15. IF CONNECTION ERROR
16. DISPLAY CONNECTION ERROR DETAILS
17. END PHP SCRIPT
18. END IF
19. CREATE 'businessID' VARIABLE FROM SESSION
20. CREATE QUERY TO SELECT EVERY LISTING THE BUSINESS HAS MADE
21. EXECUTE THE QUERY
22. IF THE QUERY IS SUCCESSFUL
23. COUNT THE NUMBER OF ROWS RETURNED
24. IF MORE THAN 0 ROWS ARE RETURNED
25. START LOOP FOR EACH ROW RETURNED
26. CREATE VARIABLES USING DATA FROM DATABASE
27. DISPLAY JOB CARD WITH JOB DETAILS AND VIEW APPLICANTS BUTTON
28. END LOOP
29. ELSE
30. DISPLAY MESSAGE SAYING BUSINESS HAS NOT POSTED ANY JOB LISTINGS YET
31. END IF
32. ELSE
33. DISPLAY QUERY ERROR MESSAGE
34. END IF
35. CLOSE DATABASE CONNECTION

2.7.12 View Applicants Page

1. START SESSION
2. IF 'logOutBtn' POST VARIABLE IS SET
3. DESTORY SESSION
4. NAVIGATE TO 'index.html'
5. END IF
6. IF 'accountID' SESSION VARIABLE IS NOT SET
7. NAVIGATE TO 'userLogin.php'
8. END IF
9. IF 'accountType' SESSION VARIABLE IS 'Personal'
10. NAVIGATE TO 'userHome.php'
11. END IF

12. SET ERROR REPORTING TO FATAL AND PARSE ONLY
13. ESTABLISH DATABASE CONNECTION VARIABLES
14. ATTEMPT TO CONNECT TO THE DATABASE
15. IF CONNECTION ERROR
16. DISPLAY CONNECTION ERROR DETAILS
17. END PHP SCRIPT
18. END IF
19. CREATE VARIABLES USING GET METHOD AND SESSION
20. CREATE QUERY TO DISPLAY JOB DETAILS
21. EXECUTE THE QUERY
22. IF THE QUERY IS SUCCESSFUL
23. STORE DETAILS IN ASSOCIATIVE ARRAY
24. CREATE VARIABLES USING DATA FROM DATABASE
25. DISPLAY JOB DETAILS
26. ELSE
27. DISPLAY QUERY ERROR MESSAGE
28. END IF
29. CREATE QUERY TO DISPLAY APPLICANT DETAILS
30. EXECUTE THE QUERY
31. IF THE QUERY IS SUCCESSFUL
32. COUNT THE NUMBER OF ROWS RETURNED
33. IF MORE THAN 0 ROWS ARE RETURNED
34. START LOOP FOR EACH ROW RETURNED
35. STORE APPLICANT DETAILS IN ASSOCIATIVE ARRAY
36. CREATE VARIABLES USING DATA FROM DATABASE
37. DISPLAY APPLICANT DETAILS
38. END LOOP
39. ELSE
40. DISPLAY MESSAGE SAYING NO APPLICANTS YET
41. END IF
42. ELSE
43. DISPLAY QUERY ERROR MESSAGE
44. ELSE IF
45. CLOSE DATABASE CONNECTION

3. Implementation

3.1 Create Database

```
CREATE DATABASE project;
USE project;

#Create table for user details
CREATE TABLE user (
  userID INT NOT NULL AUTO_INCREMENT,
  userName VARCHAR(70) NOT NULL,
  email VARCHAR(100) NOT NULL,
  password VARCHAR(30) NOT NULL CHECK(LENGTH(password) >= 8),

  PRIMARY KEY(userID)
);

#Create table for business details
CREATE TABLE business (
  businessID INT NOT NULL AUTO_INCREMENT,
  businessName VARCHAR(70) NOT NULL,
  email VARCHAR(100) NOT NULL,
  phoneNo VARCHAR(15) NOT NULL,
  password VARCHAR(30) NOT NULL CHECK(LENGTH(password) >= 8),

  PRIMARY KEY(businessID)
);

#Create table for job listings
CREATE TABLE job (
  jobID INT NOT NULL AUTO_INCREMENT,
  title VARCHAR(80) NOT NULL,
  jobType VARCHAR(14) NOT NULL CHECK(type IN('Full-time', 'Part-time', 'Apprenticeship')),
  businessID INT NOT NULL,
  description VARCHAR(4000) NOT NULL,
  weeklyHours DECIMAL(3,2) NOT NULL CHECK(weeklyHours <= 168),
  hourlyRate DECIMAL(5,2) NOT NULL,
  location VARCHAR(200) NOT NULL,

  PRIMARY KEY(jobID),
  FOREIGN KEY(businessID) REFERENCES business(businessID)
);

#Create table for applications
CREATE TABLE application (
  applicationID INT NOT NULL AUTO_INCREMENT,
  userID INT NOT NULL,
  jobID INT NOT NULL,
  coverLetter VARCHAR(2000),
  PRIMARY KEY(applicationID),
  FOREIGN KEY(userID) REFERENCES user(userID),
  FOREIGN KEY(jobID) REFERENCES job(jobID)
);

#Insert into user table
```

```
INSERT INTO user VALUES (1,'a','a@a.com','a'),(2,'John Smith','johnsmith@email.com','JSmith24'),(3,'Kara White','karawhite@email.com','Password1'),(4,'Sam Richardon','samrich03@email.com','SamRich123'),(5,'Ameerah Kiani','ak2006@email.com','Password1'),(6,'Emma Taylor','fashionexplorer23@email.com','FdcDf3fs9'),(7,'Alex Johnson','techuser23@example.com','TRf23SA23!'),(8,'Alex Thompson','avid.gamer.87@email.com','GameOn#2024'),(9,'Sarah Smith','sara.smith@email.com','sarasmith123');
```

#Insert into business table

```
INSERT INTO business VALUES (1,'b','b@b.com','123','b'),(2,'Fresh Fruits Ltd','fresh@fruits.com','01632 963847','FreshFruits'),(3,'Vintage Clothes','contact@vintageclothes.com','01632 963826','VintageClothes'),(4,'Artistic Co','sales@artisticco.com','01632 963745','ArtisticCo'),(5,'HorizonTech Innovations','tech@horizon.com','01632 969821','HorizonTech'),(6,'Spice Harbour','restaurant@spiceharbour.com','01632 960935','in07Y3xhKN'),(7,'Style Studio','enquiries@stylestudio.com','01632 968462','XA0x8iq86S'),(8,'Urban Elegance','elegance@urbanshop.com','01632 963847','U75aOlPe1n'),(9,'Pixel Playgrounds','playgrounds@pixelstudio.com','01632 963746','hb39H98Q60'),(10,'MasterCraft Construction','construction@mastercraft.com','01632 962846','QR9eb4mT3W'),(11,'QuantumLeap Innovators','innovation@quantumleap.com','01632 963821','Quantum2024'),(12,'QuantumCraft Studios','studios@quantum.com','01632 385739','QuantumStudios');
```

#Insert into job table

```
INSERT INTO `job` VALUES (1,'Team Member','Full-time',2,'Job Description',35.00,11.50,'Glasgow'),(2,'Sales Assistant','Part-time',3,'Job Description',14.00,10.42,'Paisley'),(3,'Device Expert','Part-time',5,'Job Description',12.00,12.00,'Renfrew'),(4,'Shift Manager','Full-time',8,'Job Description',38.00,12.50,'Renfrew'),(5,'Sales Assistant - Full Time','Full-time',8,'Job Description',36.00,11.80,'Renfrew'),(6,'Sales Assistant - Part Time','Part-time',8,'Job Description',20.00,11.80,'Renfrew'),(7,'Head Chef','Full-time',6,'Job Description',40.00,13.70,'Paisley'),(8,'Front of House','Part-time',6,'Job Description',18.00,9.80,'Paisley'),(9,'Games Developer','Full-time',9,'Job Description',35.00,10.42,'Glasgow'),(10,'Warehouse Operator','Full-time',8,'Job Description',40.00,12.00,'Renfrew'),(11,'Warehouse Manager','Full-time',8,'Job Description',40.00,13.50,'Renfrew'),(12,'Bricklayer','Apprenticeship',10,'Job Description',18.00,5.80,'Paisley'),(13,'Software Engineer','Full-time',12,'Job Description',40.00,14.00,'Glasgow');
```

#Insert into application table

```
INSERT INTO application VALUES (1,2,3,'Cover Letter'),(2,4,2,'Cover Letter'),(3,4,6,'Cover Letter'),(4,4,7,'Cover Letter'),(5,5,9,'Cover Letter'),(6,4,10,'Cover Letter'),(7,3,10,'Cover Letter'),(8,4,12,'Cover Letter');
```

Cover letters and job descriptions have not been included in the SQL script as they are too long.

3.2 Create Pages

3.2.1 Landing Page

index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>Landing Page</title>
    <link rel="stylesheet" type="text/css" href="styles.css">
    <link rel="icon" type="image/x-icon" href="images/favicon.png">
  </head>
```

```

<body>
  <header>
    
  </header>

  <nav>
    <ul>
      <li><a href="userLogin.php" class="right">Login / Sign Up</a></li>
      <li><a href="cvguide.php" class="right">CV Guide</a></li>
    </ul>
  </nav>

  <main>
    <section class="content">
      <h2>Why is a good CV important?</h2>
      <p>Your CV is the first impression potential employers will have of you. Your CV demonstrates professionalism and highlights your key skills and experience. It is vital to make a positive first impression which clearly outlines why you are the correct person for the job and a good CV can significantly increase your chances of being considered for the job opportunities.</p>
    </section>

    <section class="content flex">
      <section class="flexChild centerText">
        <h2>CV Guide</h2>
        <p>View our full guide on how to write a professional CV.</p>
        <button type="button" onclick="window.location.href='cvguide.php'">View
Now</button>
      </section>
      <section class="flexChild centerText">
        
      </section>
    </section>
  </main>
</body>
</html>

```

3.2.2 CV Guide Page

```

cvguide.php
<?php
  // Start session
  session_start();
?>
<!DOCTYPE html>
<html>
  <head>
    <title>CV Guide</title>

```

```

<link rel="stylesheet" type="text/css" href="styles.css">
<link rel="icon" type="image/x-icon" href="images/favicon.png">
</head>

<body>
  <header>
    
  </header>

  <nav>
    <ul>
      <?php
        if (isset($_SESSION['accountID'])) {
          echo "<ul>
            <li><a href='userHome.php' class='left'>Home</a></li>
            <li><a href='jobSearch.php' class='left'>Job Search</a></li>
            <li><a href='applications.php' class='left'>Applications</a></li>

            <li class='dropdown right'>
              <a href='javascript:void(0)' class='dropdownBtn'>Profile ></a>
              <div class='dropdown-content'>
                <a href='profile.php'>View Profile</a>
                <a id='dropdownBottom' href=''>
                  <form method='POST'>
                    <input type='submit' id='logOutButton' name='logOutBtn' value='Log
Out'>
                  </form>
                </a>
              </div>
            </li>
          </ul>";
        }
        else {
          echo "<li><a href='userLogin.php' class='right'>Login / Sign Up</a></li>
            <li><a href='index.html' class='right'>Landing Page</a></li>";
        }
      ?>
    </ul>
  </nav>

  <main>
    <section class="content">
      <h2>What makes a good CV?</h2>
      <p>Your CV should present your qualifications, skills and experience to potential employers. Some key things to include are listed below.</p>
      <p>Remember your CV should be formatted so that it is easy to read and pleasant to look at. It should be tailored for each job and should be error free.</p>
      <h3>1. Contact Information</h3>
      <p>Include your full name, phone number and email adress. You could also include a LinkedIn profile if you have one.</p>
      <h3>2. Education</h3>

```

```

    <p>What qualifications have you achieved? Are you working towards any further
    qualifications? Organise these in a list starting with the most recent.</p>
    <h3>3. Experience</h3>
    <p>What previous job roles have you had? Describe your major accomplishments and
    achievements in these roles. Organise these in a list starting with the most recent.</p>
    <h3>4. Skills</h3>
    <p>What skills do you have that are relevant to the position you are applying for? You
    should provide examples of how you can apply these skills.</p>
    <h3>5. Achievements</h3>
    <p>Have you recieved any significant achievements throughout your academic and
    professional careers?</p>
    <h3>6. References</h3>
    <p>Include some references or indicate that they are available upon request. Make sure
    you inform your references in advance.</p>
    <p>Some people you could use as referecnes are, but not limited to, <b>teachers and
    lecturers</b>, <b>managers and supervisors</b>, and <b>colleagues or team members</b>.</p>
    <h3>7. Other</h3>
    <p>You may also wish to include <b>certificates and training</b> which are relavant to
    the role, <b>languages</b> that you speak and your level of fluency, <b>projects</b> you have
    worked on including your role within the project, and <b>publications and presentations</b> you
    have contributed to.</p>
  </section>

  <section class="content">
    <h2>Want some more help?</h2>
    <p>Here are some useful external resources:</p>
    <section class="links">
      <a href="https://nationalcareers.service.gov.uk/careers-advice/cv-sections"
      target="_blank"><button type="button">UK Government</button></a>
      <a
      href="https://www.indeed.com/career-advice/resumes-cover-letters/how-to-write-a-cv"
      target="_blank"><button type="button">Indeed</button></a>
    </section>
  </section>

  <br>
</main>
</body>
</html>

```

3.2.3 Login Page

User Login

```

userLogin.php
<?php
  //Start session
  session_start();
?>
<!DOCTYPE html>

```

```

<html>
  <head>
    <title>User Login</title>
    <link rel="stylesheet" type="text/css" href="styles.css">
    <link rel="icon" type="image/x-icon" href="images/favicon.png">
  </head>

  <body>
    <section class="centerBox" id="loginBox">
      <ul id="tabSelect">
        <li><a href="userLogin.php" class="active" style="border-top-left-radius:
16px;">Personal</a></li>
        <li><a href="businessLogin.php" style="border-top-right-radius: 16px;">Business</a></li>
      </ul>

      <section class="loginForm">
        <section class="centerText">
          
        </section>
        <h1 class="centerText">Log In</h1>
        <p class="centerText">PERSONAL ACCOUNT</p>

        <?php
          // Set error reporting to exclude warnings
          // Only fatal errors and parse errors are displayed
          error_reporting(E_ERROR | E_PARSE);

          if (ISSET($_SESSION['newAccount'])) {
            // New account display message to tell user to log in
            echo "<p><b>You're almost there! Please sign into your new account.</b></p>";
            UNSET($_SESSION['newAccount']);
          }
        ?>

        <form id="userLoginForm" action="userLogin.php" method="POST">
          <label for="email">Email*</label><br>
          <input type="text" id="email" name="userEmail" required maxlength="100"><br><br>
          <label for="pword">Password*</label><br>
          <input type="password" id="pword" name="password" required
maxlength="30"><br><br>
          <section class="flex">
            <a class="flexChild50" href="userSignUp.php">Create Account</a>
            <input class="flexChild50" value="Log In" type="submit" name="loginBtn">
          </section>
        </form>
        <br>
      </section>
    </section>

    <?php
      if (ISSET($_SESSION['accountID'])) {

```

```

// User has already logged in, navigate to home page
if ($_SESSION['accountType'] == "Personal") {
    // Account type is personal so go to userHome.php
    header("Location:userHome.php");
}
elseif ($_SESSION['accountType'] == "Business") {
    // Account type is business so go to businessHome.php
    header("Location:businessHome.php");
}
}

if (isset($_POST['logInBtn'])) {
    // Database connection variables are established
    $dbHost = "localhost";
    $dbUser = "root";
    $dbPassword = "";
    $dbName = "project";

    // Attempt to connect to database
    $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

    // Check if the connection is successful, else display appropriate error message
    if (mysqli_connect_errno()) {
        echo "<h2>Connection Error</h2>";
        // Display connection error message
        echo mysqli_connect_error();
        die();
    }

    // Variables are created with the data input on the HTML form
    $email = $_POST['userEmail'];
    $password = $_POST['password'];

    // Create query to search database with account details entered
    $query = "SELECT * FROM user WHERE user.email = '$email' AND user.password = '$password'";

    // Execute the query using the active database
    $result = mysqli_query($connection, $query);

    // Check whether query has been successful
    if ($result) {
        // Count number of rows, if 1 row is returned the details are valid
        $row_count = mysqli_num_rows($result);

        if ($row_count == 1) {
            // Assign the userID and the account type to session variables
            $row = mysqli_fetch_assoc($result);
            $_SESSION['accountID'] = $row['userID'];
            $_SESSION['accountType'] = "Personal";
        }
    }
}

```

```

        // If a row is returned the details are correct, navigate to userHome.php
        header("Location:userHome.php");
    }
    else {
        // If no records are returned then details are incorrect or account type is business
        echo "<p>Could not find your account, try:</p>";
        <ol>
            <li><a href='businessLogin.php'>Log in as a business instead</a></li>
            <li><a href='userSignUp.php'>Create a new account</a></li>
        </ol>";
    }
}
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

// Close the database connection
mysqli_close($connection);
}
?>
</body>
</html>

```

Business Login

businessLogin.php

```

<?php
    //Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Business Login</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <section class="centerBox" id="loginBox">
            <ul id="tabSelect">
                <li><a href="userLogin.php" style="border-top-left-radius: 16px;">Personal</a></li>
                <li><a href="businessLogin.php" class="active" style="border-top-right-radius:
16px;">Business</a></li>
            </ul>

            <section class="loginForm">
                <section class="centerText">
                    

```

```

</section>
<h1 class="centerText">Log In</h1>
<p class="centerText">BUSINESS ACCOUNT</p>

<?php
    // Set error reporting to exclude warnings
    // Only fatal errors and parse errors are displayed
    error_reporting(E_ERROR | E_PARSE);

    if (ISSET($_SESSION['newAccount'])) {
        // New account display message to tell user to log in
        echo "<p><b>You're almost there! Please sign into your new account.</b></p>";
        UNSET($_SESSION['newAccount']);
    }
?>

<form id="businessLoginForm" action="businessLogin.php" method="POST">
    <label for="email">Email*</label><br>
    <input type="text" id="email" name="businessEmail" required
maxlength="100"><br><br>
    <label for="pword">Password*</label><br>
    <input type="password" id="pword" name="password" required
maxlength="30"><br><br>

    <section class="flex">
        <a class="flexChild50" href="businessSignUp.php">Create Account</a>
        <input class="flexChild50" value="Log In" type="submit" name="logInBtn">
    </section>
</form>
<br>
</section>
</section>

<?php
    if (ISSET($_SESSION['accountID'])) {
        // User has already logged in, navigate to home page
        if ($_SESSION['accountType'] == "Personal") {
            // Account type is personal so go to userHome.php
            header("Location:userHome.php");
        }
        elseif ($_SESSION['accountType'] == "Business") {
            // Account type is business so go to businessHome.php
            header("Location:businessHome.php");
        }
    }

    if (ISSET($_POST['logInBtn'])) {
        // Database connection variables are established
        $dbHost = "localhost";
        $dbUser = "root";
        $dbPassword = "";
        $dbName = "project";
    }
}

```

```

// Attempt to connect to database
$connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

// Check if the connection is successful, else display appropriate error message
if (mysqli_connect_errno()) {
    echo "<h2>Connection Error</h2>";
    // Display connection error message
    echo mysqli_connect_error();
    die();
}

// Variables are created with the data input on the HTML form
$email = $_POST['businessEmail'];
$password = $_POST['password'];

// Create query to search database with account details entered
$query = "SELECT * FROM business WHERE business.email = '$email' AND
business.password = '$password'";

// Execute the query using the active database
$result = mysqli_query($connection, $query);

// Check whether query has been succesful
if ($result) {
    // Count number of rows, if 1 row is returned the details are valid
    $row_count = mysqli_num_rows($result);

    if ($row_count == 1) {
        // Assign the businessID and the account type to session variables
        $row = mysqli_fetch_assoc($result);
        $_SESSION['accountID'] = $row['businessID'];
        $_SESSION['accountType'] = "Business";

        // If a row is returned the details are correct, navigate to businessHome.php
        header("Location:businessHome.php");
    }
    else {
        // If no records are returned then details are incorrect or account type is personal
        echo "<p>Could not find your account, try:</p>";
        <ol>
            <li><a href='userLogin.php'>Log in as a user instead</a></li>
            <li><a href='businessSignUp.php'>Create a new account</a></li>
        </ol>";
    }
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

```

```

        // Close the database connection
        mysqli_close($connection);
    }
    ?>
</body>
</html>

```

3.2.4 Sign Up Page

User Sign Up

userSignUp.php

```

<?php
    //Start session
    session_start();
    ?>
<!DOCTYPE html>
<html>
    <head>
        <title>User Sign Up</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <section class="centerBox" id="loginBox">
            <ul id="tabSelect">
                <li><a href="userSignUp.php" class="active" style="border-top-left-radius:
16px;">Personal</a></li>
                <li><a href="businessSignUp.php" style="border-top-right-radius:
16px;">Business</a></li>
            </ul>

            <section class="loginForm">
                <section class="centerText">
                    
                </section>
                <h1 class="centerText">Sign Up</h1>
                <p class="centerText">PERSONAL ACCOUNT</p>

                <form id="userSignUpForm" action="userSignUp.php" method="POST">
                    <label for="uname">Name*</label><br>
                    <input type="text" id="uname" name="userName" required maxlength="70"><br><br>
                    <label for="email">Email*</label><br>
                    <input type="email" id="email" name="userEmail" required
maxlength="100"><br><br>
                    <label for="pword"> Password*</label><br>
                    <input type="password" id="pword" name="password" required minlength="8"
maxlength="30"><br><br>

```

```

        <section class="centerText">
            <input type="submit" value="Create Account" style="width: 50%;"
name="signUpBtn">
        </section>
    </form>
</section>
</section>

<?php
    if (ISSET($_SESSION['accountID'])) {
        // User has already logged in, navigate to home page
        if ($_SESSION['accountType'] == "Personal") {
            // Account type is personal so go to userHome.php
            header("Location:userHome.php");
        }
        elseif ($_SESSION['accountType'] == "Business") {
            // Account type is business so go to businessHome.php
            header("Location:businessHome.php");
        }
    }
    if (ISSET($_POST['signUpBtn'])) {
        // Set error reporting to exclude warnings
        // Only fatal errors and parse errors are displayed
        error_reporting(E_ERROR | E_PARSE);

        // Database connection variables are established
        $dbHost = "localhost";
        $dbUser = "root";
        $dbPassword = "";
        $dbName = "project";

        // Attempt to connect to database
        $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

        // Check if the connection is successful, else display appropriate error message
        if (mysqli_connect_errno()) {
            echo "<h2>Connection Error</h2>";
            // Display connection error message
            echo mysqli_connect_error();
            die();
        }

        // Variables are created with the data input on the HTML form
        $uname = $_POST['userName'];
        $email = $_POST['userEmail'];
        $pwd = $_POST['password'];

        // Create a query to check if the email address has already been used
        $query = "SELECT * FROM user WHERE user.email = '$email'";

        //Execute the query using the active database
    }

```

```

$result = mysqli_query($connection, $query);

// Check whether the query has been successful and if email has been used previously
if ($result) {
    // Count the number rows, if 0 rows are returned then the email has not been used
already
    $row_count = mysqli_num_rows($result);

    if ($row_count == 0) {
        // Email has not already been used and a new account can be created
        // Create a query to create a new account
        $query = "INSERT INTO user(userName, email, password) VALUES('$uname', '$email',
'$pwd')";

        // Execute the query using the active database
        $result = mysqli_query($connection, $query);

        // Check query was successful
        if ($result) {
            // Set newAccount session variable and navigate to userLogin.php
            $_SESSION['newAccount'] = "True";
            header("location:userLogin.php");
        }
        else {
            // If results couldn't be returned display appropriate error message
            echo "<p>Error creating your account</p>";
            echo "<p>Error description: " . mysqli_error($connection) . "</p>";
        }
    }
    else {
        // The email has already been used, display error message to user
        echo "<p>This email is already in use, try <a href='userLogin.php'>logging in</a>
instead?</p>";
    }
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

// Close the database connection
mysqli_close($connection);
}
?>
</body>
</html>

```

businessSignUp.php

```

<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Business Sign Up</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <section class="centerBox" id="loginBox">
            <ul id="tabSelect">
                <li><a href="userSignUp.php" style="border-top-left-radius: 16px;">Personal</a></li>
                <li><a href="businessSignUp.php" class="active" style="border-top-right-radius:
16px;">Business</a></li>
            </ul>

            <section class="loginForm">
                <section class="centerText">
                    
                </section>
                <h1 class="centerText">Sign Up</h1>
                <p class="centerText">BUSINESS ACCOUNT</p>

                <form id="businessSignUpForm" action="businessSignUp.php" method="POST">
                    <label for="bname">Business Name*</label><br>
                    <input type="text" id="bname" name="businessName" required
maxlength="70"><br><br>
                    <label for="email">Email*</label><br>
                    <input type="email" id="email" name="businessEmail" required
maxlength="100"><br><br>
                    <label for="phoneNo">Phone Number*</label><br>
                    <input type="text" id="phoneNo" name="phoneNumber" required
maxlength="15"><br><br>
                    <label for="pword">Password*</labe><br>
                    <input type="password" id="pword" name="password" required minlength="8"
maxlength="30"><br><br>
                    <section class="centerText">
                        <input type="submit" value="Create Account" style="width: 50%;"
name="signUpBtn">
                    </section>
                </form>
            </section>
        </section>
    </body>
</html>

```

```

<?php
    if (ISSET($_SESSION['accountID'])) {
        // User has already logged in, navigate to home page
        if ($_SESSION['accountType'] == "Personal") {
            // Account type is personal so go to userHome.php
            header("Location:userHome.php");
        }
        elseif ($_SESSION['accountType'] == "Business") {
            // Account type is business so go to businessHome.php
            header("Location:businessHome.php");
        }
    }
    if (ISSET($_POST['signUpBtn'])) {
        // Set error reporting to exclude warnings
        // Only fatal errors and parse errors are displayed
        error_reporting(E_ERROR | E_PARSE);

        // Database connection variables are established
        $dbHost = "localhost";
        $dbUser = "root";
        $dbPassword = "";
        $dbName = "project";

        // Attempt to connect to database
        $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

        // Check if the connection is successful, else display appropriate error message
        if (mysqli_connect_errno()) {
            echo "<h2>Connection Error</h2>";
            // Display connection error message
            echo mysqli_connect_error();
            die();
        }

        // Variables are created with the data input on the HTML form
        $bname = $_POST['businessName'];
        $email = $_POST['businessEmail'];
        $phoneNo = $_POST['phoneNumber'];
        $pword = $_POST['password'];

        // Create a query to check if the email address has already been used
        $query = "SELECT * FROM business WHERE business.email = '$email'";

        //Execute the query using the active database
        $result = mysqli_query($connection, $query);

        // Check whether the query has been successful and if email has been used previously
        if ($result) {
            // Count the number rows, if 0 rows are returned then the email has not been used
            already
            $row_count = mysqli_num_rows($result);

```

```

        if ($row_count == 0) {
            // Email has not already been used and a new account can be created
            // Create a query to create a new account
            $query = "INSERT INTO business(businessName, email, phoneNo, password)
VALUES('$bname', '$email', '$phoneNo', '$pword')";

            // Execute the query using the active database
            $result = mysqli_query($connection, $query);

            // Check query was successful
            if ($result) {
                // Set newAccount session variable and navigate to businessLogin.php
                $_SESSION['newAccount'] = "True";
                header("location:businessLogin.php");
            }
            else {
                // If results couldn't be returned display appropriate error message
                echo "<p>Error creating your account</p>";
                echo "<p>Error description: " . mysqli_error($connection) . "</p>";
            }
        }
        else {
            // The email has already been used, display error message to user
            echo "<p>This email is already in use, try <a href='businessLogin.php'>logging in</a>
instead?</p>";
        }
    }
    else {
        // If results couldn't be returned display appropriate error message
        echo "<p>Error searching database</p>";
        echo "<p>Error description: " . mysqli_error($connection) . "</p>";
    }

    // Close the database connection
    mysqli_close($connection);
}
?>
</body>
</html>

```

3.2.5 User Home Page

userHome.php

```

<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>

```

```

<head>
  <title>Home</title>
  <link rel="stylesheet" type="text/css" href="styles.css">
  <link rel="icon" type="image/x-icon" href="images/favicon.png">
</head>

<body>
  <header>
    
  </header>

  <nav>
    <ul>
      <li><a href="userHome.php" class="left">Home</a></li>
      <li><a href="jobSearch.php" class="left">Job Search</a></li>
      <li><a href="applications.php" class="left">Applications</a></li>

      <li class="dropdown right">
        <a href="javascript:void(0)" class="dropdownBtn">Profile ></a>
        <div class="dropdown-content">
          <a href="profile.php">View Profile</a>
          <a id="dropdownBottom" href="">
            <form method="POST">
              <input type="submit" id="logOutButton" name="logOutBtn" value="Log Out">
            </form>
          </a>
        </div>
      </li>
    </ul>

    <?php
      if (isset($_POST['logOutBtn'])) {
        // Log out button has been pressed, destroy session and navigate to landing page
        session_destroy();
        header("Location:index.html");
      }
    ?>
  </nav>

  <main>
    <section class="flex">
      <section class="content flexChild75">
        <h2>Recent Job Postings</h2>

        <?php
          if (!isset($_SESSION['accountID'])) {
            // User has not logged in, navigate to userLogin.php
            header("Location:userLogin.php");
          }

          // Check if the account type is business

```

```

if ($_SESSION['accountType'] == "Business") {
    // Account type is business, navigate to businessHome.php
    header("Location:businessHome.php");
}

// User has logged in, create query to get 3 most recent job listings
// Set error reporting to exclude warnings
// Only fatal errors and parse errors are displayed
error_reporting(E_ERROR | E_PARSE);

// Database connection variables are established
$dbHost = "localhost";
$dbUser = "root";
$dbPassword = "";
$dbName = "project";

// Attempt to connect to database
$connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

// Check if the connection is successful, else display appropriate error message
if (mysqli_connect_errno()) {
    echo "<h2>Connection Error</h2>";
    // Display connection error message
    echo mysqli_connect_error();
    die();
}

// Create query to get 3 most recent jobs
$query = "SELECT * FROM job ORDER BY jobID DESC LIMIT 3";

// Execute the query using the active database
$result = mysqli_query($connection, $query);

// Check whether query has been succesful
if ($result) {
    // Count number of rows, if 1 or more are returned display most recent jobs
    $row_count = mysqli_num_rows($result);

    if ($row_count > 0) {
        // Display most recent jobs in a section
        while ($row = mysqli_fetch_assoc($result)) {
            // Declare variables from database
            $jobID = $row['jobID'];
            $jobTitle = $row['title'];
            $briefDesc = htmlspecialchars_decode(substr($row['description'], 0, 100) .
" ...");

            echo "<section class='flex card'>
                <section class='flexChild75'>
                    <h3>$jobTitle</h3>
                    <p>$briefDesc</p>

```

```

        </section>
        <section class='flexchild25 centerText flexCenter'>
            <a href='apply.php?job=$jobID'><button type='button'>Apply
Now</button></a>
        </section>
    </section>";
    }
}
else {
    // If no records are returned then no jobs were found
    echo "<p>No jobs were found in the database. Try refreshing this page.</p>";
}
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

// Close the database connection
mysqli_close($connection);
?>
</section>
<section class="content flexChild25">
    <h2>CV Guide</h2>
    <section class="centerText">
        
        <p>View our full guide on how to write a professional CV to maximise the chances of
being noticed by employers.</p>
        <button type="button" onclick="window.location.href='cvguide.php'">View
Now</button>
    </section>
</section>
</main>
</body>
</html>

```

3.2.6 Job Search Page

```

jobSearch.php
<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Job Search</title>
        <link rel="stylesheet" type="text/css" href="styles.css">

```

```

<link rel="icon" type="image/x-icon" href="images/favicon.png">
</head>

<body>
  <header>
    
  </header>

  <nav>
    <ul>
      <li><a href="userHome.php" class="left">Home</a></li>
      <li><a href="jobSearch.php" class="left">Job Search</a></li>
      <li><a href="applications.php" class="left">Applications</a></li>

      <li class="dropdown right">
        <a href="javascript:void(0)" class="dropdownBtn">Profile ></a>
        <div class="dropdown-content">
          <a href="profile.php">View Profile</a>
          <a id="dropdownBottom" href="">
            <form method="POST">
              <input type="submit" id="logOutButton" name="logOutBtn" value="Log Out">
            </form>
          </a>
        </div>
      </li>
    </ul>

    <?php
      if (isset($_POST['logOutBtn'])) {
        // Log out button has been pressed, destroy session and navigate to landing page
        session_destroy();
        header("Location:index.html");
      }
    ?>
  </nav>

  <main>
    <section class="flex">
      <section class="content flexChild25" style="max-height: 470px;">
        <h2>Filter</h2>
        <form action="jobSearch.php" method="POST" id="jobSearchForm">
          <input type="text" name="search" id="searchfield" placeholder="Search by title or
location..."><br><br>

          <label>Job type</label><br>
          <input type="radio" id="partTime" name="jobType" value="Part-time" checked>
          <label for="partTime">Part-time</label><br>
          <input type="radio" id="fullTime" name="jobType" value="Full-time">
          <label for="fullTime">Full-time</label><br>
          <input type="radio" id="apprenticeship" name="jobType" value="Apprenticeship">
          <label for="apprenticeship">Apprenticeship</label><br><br>

```

```

<label for="wkHrs">Weekly hours</label><br>
<select name="weeklyHours" id="wkHrs">
  <option value="0">Any</option>
  <option value="5">5 hours +</option>
  <option value="10">10 hours +</option>
  <option value="15">15 hours +</option>
  <option value="20">20 hours +</option>
  <option value="25">25 hours +</option>
  <option value="30">30 hours +</option>
</select><br><br>

<label for="hrRate">Hourly rate</label><br>
<select name="hourlyRate" id="hrRate">
  <option value="0">Any</option>
  <option value="5">£5/hr +</option>
  <option value="7">£7/hr +</option>
  <option value="9">£9/hr +</option>
  <option value="11">£11/hr +</option>
  <option value="13">£13/hr +</option>
</select><br><br>

<section class="centerText">
  <input type="submit" name="searchButton" id="submitBtn" value="Search">
</section><br>
</form>
</section>

<section class="content flexChild75">
  <h2>Available Jobs</h2>

  <?php
    if (!isset($_SESSION['accountID'])) {
      // User has not logged in, navigate to userLogin.php
      header("Location:userLogin.php");
    }

    // Check if the account type is business
    if ($_SESSION['accountType'] == "Business") {
      // Account type is business, navigate to businessHome.php
      header("Location:businessHome.php");
    }

    if (isset($_POST['searchButton'])) {
      // Set error reporting to exclude warnings
      // Only fatal errors and parse errors are displayed
      error_reporting(E_ERROR | E_PARSE);

      // Database connection variables are established
      $dbHost = "localhost";
      $dbUser = "root";
    }
  }

```

```

$dbPassword = "";
$dbName = "project";

// Attempt to connect to database
$connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

// Check if the connection is successful, else display appropriate error message
if (mysqli_connect_errno()) {
    echo "<h2>Connection Error</h2>";
    // Display connection error message
    echo mysqli_connect_error();
    die();
}

// Create variables with data from html form
$searchString = $_POST['search'];
$jobType = $_POST['jobType'];
$wkHours = $_POST['weeklyHours'];
$shrRate = $_POST['hourlyRate'];

// Create query to get jobs matching search options
$query = "SELECT * FROM job, business WHERE job.businessID =
business.businessID AND jobType = '$jobType' AND weeklyHours >= '$wkHours' AND hourlyRate
>= '$shrRate' AND (title LIKE '%$searchString%' OR location LIKE '%$searchString%') ORDER BY
jobID DESC";

// Execute the query using the active database
$result = mysqli_query($connection, $query);

// Check whether query has been successful
if ($result) {
    // Count number of rows, if 1 or more are returned display most recent jobs
    $row_count = mysqli_num_rows($result);
    echo "<p>$row_count results</p>";

    if ($row_count > 0) {
        // Display most recent jobs in a section
        while ($row = mysqli_fetch_assoc($result)) {
            // Declare job details variables from database
            $jobID = $row['jobID'];
            $jobTitle = $row['title'];
            $jobType = $row['jobType'];
            $briefDesc = substr($row['description'], 0, 100) . "...";
            $hourlyRate = $row['hourlyRate'];
            $wkHours = $row['weeklyHours'];
            $location = $row['location'];

            // Declare business details variables from database
            $busName = $row['businessName'];

            echo "<section class='flex card'>";

```

```

        <section class='flexChild75'>
            <h3>$jobTitle at $busName</h3>

            <table class='jobDetails'>
                <tr>
                    <td class='tdShade'>Job Type</td>
                    <td>$jobType</td>
                    <td class='tdShade'>Hourly Rate</td>
                    <td>$hourlyRate</td>
                </tr>
                <tr>
                    <td class='tdShade'>Weekly Hours</td>
                    <td>$wkHours</td>
                    <td class='tdShade'>Location</td>
                    <td>$location</td>
                </tr>
            </table>

            <p>$briefDesc</p>
        </section>
        <section class='centerText flexCenter flexChild25'>
            <a href='apply.php?job=$jobID'><button type='button'>Apply
Now</button></a>

        </section>
    </section>";
    }
}
else {
    // If no records are returned then no jobs were found
    echo "<p>No jobs were found in the database. Try altering your filters.</p>";
}
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

// Close the database connection
mysqli_close($connection);
}
else {
    // No search has been made
    echo "<p>Make a search query to start searching for jobs.</p>";
}
?>
</section>
</section>
</main>
</body>
</html>

```

3.2.7 Apply for Job Page

apply.php

```
<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Apply for Job</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <header>
            
        </header>

        <nav>
            <ul>
                <li><a href="userHome.php" class="left">Home</a></li>
                <li><a href="jobSearch.php" class="left">Job Search</a></li>
                <li><a href="applications.php" class="left">Applications</a></li>

                <li class="dropdown right">
                    <a href="javascript:void(0)" class="dropdownBtn">Profile ></a>
                    <div class="dropdown-content">
                        <a href="profile.php">View Profile</a>
                        <a id="dropdownBottom" href="">
                            <form method="POST">
                                <input type="submit" id="logOutButton" name="logOutBtn" value="Log Out">
                            </form>
                        </a>
                    </div>
                </li>
            </ul>

            <?php
                if (isset($_POST['logOutBtn'])) {
                    // Log out button has been pressed, destroy session and navigate to landing page
                    session_destroy();
                    header("Location:index.html");
                }
            ?>
        </nav>

        <main>
            <?php
                if (!isset($_SESSION['accountID'])) {
```

```

        // User has not logged in, navigate to userLogin.php
        header("Location:userLogin.php");
    }

    // Check if the account type is business
    if ($_SESSION['accountType'] == "Business") {
        // Account type is business, navigate to businessHome.php
        header("Location:businessHome.php");
    }

    // User has logged in, display job details
    // Set error reporting to exclude warnings
    // Only fatal errors and parse errors are displayed
    error_reporting(E_ERROR | E_PARSE);

    // Database connection variables are established
    $dbHost = "localhost";
    $dbUser = "root";
    $dbPassword = "";
    $dbName = "project";

    // Attempt to connect to database
    $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

    // Check if the connection is successful, else display appropriate error message
    if (mysqli_connect_errno()) {
        echo "<h2>Connection Error</h2>";
        // Display connection error message
        echo mysqli_connect_error();
        die();
    }

    // Variables are created from data sent using GET method
    $jobID = $_GET['job'];

    // Create query to get details for selected job
    $query = "SELECT * FROM job, business WHERE job.jobID = '$jobID' and job.businessID =
business.businessID";

    // Execute the query using the active database
    $result = mysqli_query($connection, $query);

    // Check whether query has been succesful
    if ($result) {
        // Count number of rows, if 1 or more are returned display most recent jobs
        $row_count = mysqli_num_rows($result);

        if ($row_count > 0) {
            // Job has been found in database, display details for job
            // Store job details in accosciative array
            $row = mysqli_fetch_assoc($result);
        }
    }

```

```

// Create variables as required
// Business details
$busName = $row['businessName'];
$busEmail = $row['email'];
$busPhone = $row['phoneNo'];

// Job Details
$jobTitle = $row['title'];
$jobType = $row['jobType'];
$jobDesc = htmlspecialchars_decode(str_replace("\n", "<br>", $row['description']));
$hourlyRate = $row['hourlyRate'];
$wkHours = $row['weeklyHours'];
$location = $row['location'];

// CODE FOR SUBMITTING AN APPLICATION
if (isset($_POST['submitCover'])) {
    // Application submit button has been pressed

    // Create variables for userID and cover letter (from POST)
    $userID = $_SESSION['accountID'];
    $coverLetter = htmlspecialchars($_POST['coverLetter']);

    // Create query to check application does not already exist
    $query = "SELECT * FROM application WHERE userID = '$userID' AND jobID =
'$jobID'";

    // Execute the query using the active database
    $result = mysqli_query($connection, $query);

    // Check whether query was successful
    if ($result) {
        // Count number of rows, if 0 rows are returned the application has not already
        been made

        $row_count = mysqli_num_rows($result);

        if ($row_count == 0) {
            // Application has not already been made so should be created
            // Create query to insert new application into database
            $query = "INSERT INTO application(userID, jobID, coverLetter)
VALUES('$userID', '$jobID', '$coverLetter')";

            // Execute the query using the active database
            $result = mysqli_query($connection, $query);

            // Check query was successful
            if ($result) {
                // Query was successful and application has been made
                echo "<section class='content'>
                <h2>Application Successful</h2>
                <p>You have applied for $jobTitle at $busName, good luck!</p>

```

```

        <p>View all of your applications <a
href='applications.php'>here</a>.</p>
        </section>";
    }
    else {
        // Query was not successful and application has not been made
        echo "<p>Error submitting your application</p>";
        echo "<p>Error description: " . mysqli_error($connection) . "</p>";
    }
}
else {
    // Application has already been made, display message to user
    echo "<section class='content'>
        <h2>Already Applied</h2>
        <p>You have already applied for $jobTitle at $busName, you cannot
apply again.</p>
        <p>View all of your applications <a
href='applications.php'>here</a>.</p>
        </section>";
    }
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}
}

// Display details
echo "<section class='content'>
    <h2>Apply for $jobTitle</h2>

    <table class='jobDetails'>
        <tr>
            <td class='tdShade'>Job Type</td>
            <td>$jobType</td>
            <td class='tdShade'>Hourly Rate</td>
            <td>$hourlyRate</td>
        </tr>
        <tr>
            <td class='tdShade'>Weekly Hours</td>
            <td>$wkHours</td>
            <td class='tdShade'>Location</td>
            <td>$location</td>
        </tr>
    </table>

    <section class='card'>
        <h3>Business Details</h3>
        <table>
            <tr>

```

```

        <td>Name</td>
        <td>$busName</td>
    </tr>
    <tr>
        <td>Email</td>
        <td>$busEmail</td>
    </tr>
    <tr>
        <td>Phone Number</td>
        <td>$busPhone</td>
    </tr>
</table>

<h3>Job Description</h3>
<p>$jobDesc</p>
</section>

<form id='coverLetterForm' action='apply.php?job=$jobID' method='POST'>
    <section class='card'>
        <h3>Cover Letter (Recommended)</h3>
        <p>This is where you can tell the employer a little bit about yourself.
Consider mentioning why you think you are ideal for the job.</p>

        <textarea name='coverLetter' id='coverTextArea'
maxlength='2000'></textarea>
        <p id='coverCharsRemaining'>2000 Characters Remaining</p>
    </section>

    <section id='submitApplication'>
        <input type='submit' value='Submit' id='submitBtn' name='submitCover'>
    </section>
</form>
</section>";
    }
    else {
        // Job was not found in database, display error message
        echo "<p>This job was not found in the database.</p>";
    }
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

// Close the database connection
mysqli_close($connection);
?>
</main>

<script>

```

```

const textarea = document.getElementById("coverTextArea");

textarea.addEventListener("input", event => {
    const target = event.currentTarget;
    const maxLength = target.getAttribute("maxlength");
    const currentLength = target.value.length;
    const charsRemaining = maxLength - currentLength;

    document.getElementById("coverCharsRemaining").innerText = charsRemaining + "
Characters Remaining";
});
</script>
</body>
</html>

```

3.2.8 View Applications Page

applications.php

```

<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Applications</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <header>
            
        </header>

        <nav>
            <ul>
                <li><a href="userHome.php" class="left">Home</a></li>
                <li><a href="jobSearch.php" class="left">Job Search</a></li>
                <li><a href="applications.php" class="left">Applications</a></li>

                <li class="dropdown right">
                    <a href="javascript:void(0)" class="dropdownBtn">Profile ></a>
                    <div class="dropdown-content">
                        <a href="profile.php">View Profile</a>
                        <a id="dropdownBottom" href="">
                            <form method="POST">
                                <input type="submit" id="logOutButton" name="logOutBtn" value="Log Out">
                            </form>
                        </a>
                    </div>
                </li>
            </ul>

```

```

        </div>
    </li>
</ul>

<?php
    if (isset($_POST['logOutBtn'])) {
        // Log out button has been pressed, destroy session and navigate to landing page
        session_destroy();
        header("Location:index.html");
    }
?>
</nav>

<main>
    <section class="content">
        <h2>Your Applications</h2>

        <?php
            if (!isset($_SESSION['accountID'])) {
                // User has not logged in, navigate to userLogin.php
                header("Location:userLogin.php");
            }

            // Check if the account type is business
            if ($_SESSION['accountType'] == "Business") {
                // Account type is business, navigate to businessHome.php
                header("Location:businessHome.php");
            }

            // User has logged in display all of their existing applications starting with most recent
            // Set error reporting to exclude warnings
            // Only fatal errors and parse errors are displayed
            error_reporting(E_ERROR | E_PARSE);

            // Database connection variables are established
            $dbHost = "localhost";
            $dbUser = "root";
            $dbPassword = "";
            $dbName = "project";

            // Attempt to connect to database
            $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

            // Check if the connection is successful, else display appropriate error message
            if (mysqli_connect_errno()) {
                echo "<h2>Connection Error</h2>";
                // Display connection error message
                echo mysqli_connect_error();
                die();
            }
        </?php>
    </section>
</main>

```

```

// Create variables for userID from session
$userID = $_SESSION['accountID'];

// Create query to get all existing applications for the current user
$query = "SELECT * FROM application, job, business WHERE userID = '$userID' AND
application.jobID = job.jobID AND job.businessID = business.businessID ORDER BY applicationID
DESC";

// Execute the query using the active database
$result = mysqli_query($connection, $query);

// Check whether query has been succesful
if ($result) {
    // Count number of rows, if 1 or more are returned display existing applications
    $row_count = mysqli_num_rows($result);

    if ($row_count > 0) {
        // Store result in associative array
        while ($row = mysqli_fetch_assoc($result)) {
            // Declare variables from database
            $jobTitle = $row['title'];
            $busName = $row['businessName'];
            $description = str_replace("\n", "<br>", $row['description']);
            $coverLetter = htmlspecialchars_decode($row['coverLetter']);

            // If the cover letter is blank add message for user
            if ($coverLetter == "") {
                $coverLetter = "No cover letter was provided.";
            }

            // Display job card
            echo "<section class='card'>
                <h3>$jobTitle at $busName</h3>
                <details>
                    <summary>Job Description</summary>
                    <blockquote>$description</blockquote>
                </details>
                <details>
                    <summary>Your cover letter</summary>
                    <blockquote>$coverLetter</blockquote>
                </details>
                <br>
            </section>";
        }
    }
    else {
        // If no records are returned then no jobs were found
        echo "<p>You have not applied to any jobs yet. You can search for jobs <a
href='jobSearch.php'>here</a>.</p>";
    }
}

```

```

        else {
            // If results couldn't be returned display appropriate error message
            echo "<p>Error searching database</p>";
            echo "<p>Error description: " . mysqli_error($connection) . "</p>";
        }

        // Close the database connection
        mysqli_close($connection);
    ?>
</section>
</main>
</body>
</html>

```

3.2.9 Profile Page

```

profile.php
<?php
    // start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Profile</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <?php
            if (ISSET($_SESSION['accountID'])) {
                // Set error reporting to exclude warnings
                // Only fatal errors and parse errors are displayed
                error_reporting(E_ERROR | E_PARSE);

                if (ISSET($_POST['logOutBtn'])) {
                    // Log out button has been pressed, destroy session and navigate to landing page
                    session_destroy();
                    header("Location:index.html");
                }
            }
            else {
                // Log out button has not been pressed, connect to database
                // Database connection variables are established
                $dbHost = "localhost";
                $dbUser = "root";
                $dbPassword = "";
                $dbName = "project";

                // Attempt to connect to database
            }
        }
    </body>
</html>

```

```

        $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

        // Check if the connection is successful, else display appropriate error message
        if (mysqli_connect_errno()) {
            echo "<h2>Connection Error</h2>";
            // Display connection error message
            echo mysqli_connect_error();
            die();
        }

        // Get accountID from session variable
        $accountID = $_SESSION['accountID'];
    }
}
else {
    // User has not logged in, navigate to userLogin.php
    header("Location:userLogin.php");
}
?>

<header>
    <?php
        if ($_SESSION['accountType'] == "Personal") {
            echo "<img src='images/banner.png' alt='Banner Image'>";
        } else if ($_SESSION['accountType'] == "Business") {
            echo "<img src='images/businessBanner.png' alt='Banner Image'>";
        }
    ?>
</header>

<nav>
    <?php
        if ($_SESSION['accountType'] == "Personal") {
            // Account type is personal, display nav bar for personal accounts
            echo "<ul>
                <li><a href='userHome.php' class='left'>Home</a></li>
                <li><a href='jobSearch.php' class='left'>Job Search</a></li>
                <li><a href='applications.php' class='left'>Applications</a></li>

                <li class='dropdown right'>
                    <a href='javascript:void(0)' class='dropdownBtn'>Profile ></a>
                    <div class='dropdown-content'>
                        <a href='profile.php'>View Profile</a>
                        <a id='dropdownBottom' href=''>
                            <form method='POST'>
                                <input type='submit' id='logOutButton' name='logOutBtn' value='Log
Out'>
                                </form>
                            </a>
                        </div>
                    </li>
            </ul>
        }
    ?>

```

```

        </ul>";
    }
    elseif ($_SESSION['accountType'] == "Business") {
        // Account type is business, display nav bar for business accounts
        echo "<ul>
            <li><a href='businessHome.php' class='left'>Home</a></li>
            <li><a href='createJob.php' class='left'>Create Job</a></li>
            <li><a href='viewListings.php' class='left'>View Listings</a></li>

            <li class='dropdown right'>
                <a href='javascript:void(0)' class='dropdownBtn'>Profile ></a>
                <div class='dropdown-content'>
                    <a href='profile.php'>View Profile</a>
                    <a id='dropdownBottom' href=''>
                        <form method='POST'>
                            <input type='submit' id='logOutButton' name='logOutBtn' value='Log
Out'>
                                </form>
                            </a>
                        </div>
                    </li>
                </ul>";
    }
    ?>
</nav>

<main>
<?php
    if ($_SESSION['accountType'] == "Personal") {
        // Account type is personal display section containing personal account details
        // Create query to search user table
        $query = "SELECT * FROM user WHERE user.userID = '$accountID'";

        // Execute the query using the active database
        $result = mysqli_query($connection, $query);

        // Check whether the query has been successful and display user account details
        if ($result) {
            // Count number of rows, if 1 row is returned the user exists and the details should
            be displayed
            $row_count = mysqli_num_rows($result);

            if ($row_count == 1) {
                // Get account details using associative array
                $row = mysqli_fetch_assoc($result);
                $name = $row['userName'];
                $email = $row['email'];

                // Display account details
                echo "<section class='content'>
                    <h2>Your Profile</h2>

```

```

        <p>PERSONAL ACCOUNT</p>
        <table>
            <tr>
                <td>Name</td>
                <td>$name</td>
            </tr>
            <tr>
                <td>Email</td>
                <td>$email</td>
            </tr>
        </table>
        <br>
        <form method='POST'>
            <input name='logOutBtn' type='submit' value='Log Out'>
        </form>
    </section>";
}
else {
    // No records have matched (should never happen) so display error message
    echo "<p>Could not find user in database.</p>";
}
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}
}

elseif ($_SESSION['accountType'] == "Business") {
    // Account type is business display section containing business account details
    // Create query to search business table
    $query = "SELECT * FROM business WHERE business.businessID = '$accountId'";

    // Execute the query using the active database
    $result = mysqli_query($connection, $query);

    // Check whether the query has been successful and display business account details
    if ($result) {
        // Count number of rows, if 1 row is returned the business exists and the details
        // should be displayed
        $row_count = mysqli_num_rows($result);

        if ($row_count == 1) {
            // Get account details using associative array
            $row = mysqli_fetch_assoc($result);
            $name = $row['businessName'];
            $email = $row['email'];
            $phoneNo = $row['phoneNo'];

            // Display account details

```

```

        echo "<section class='content'>
            <h2>Your Profile</h2>
            <p>BUSINESS ACCOUNT</p>
            <table>
                <tr>
                    <td>Name</td>
                    <td style='padding-left: 20px;'>{$name}</td>
                </tr>
                <tr>
                    <td>Email</td>
                    <td style='padding-left: 20px;'>{$email}</td>
                </tr>
                <tr>
                    <td>Phone Number</td>
                    <td style='padding-left: 20px;'>{$phoneNo}</td>
                </tr>
            </table>
            <br>
            <form method='POST'>
                <input name='logOutBtn' type='submit' value='Log Out'>
            </form>
        </section>";
    }
    else {
        // No records have matched (should never happen) so display error message
        echo "<p>Could not find business in database.</p>";
    }
}
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}
}
?>
</main>

<?php
    // Close the database connection
    mysqli_close($connection);
?>
</body>
</html>

```

3.2.10 Business Home Page

businessHome.php

```

<?php
    // Start session
    session_start();

```

```

?>
<!DOCTYPE html>
<html>
  <head>
    <title>Home</title>
    <link rel="stylesheet" type="text/css" href="styles.css">
    <link rel="icon" type="image/x-icon" href="images/favicon.png">
  </head>

  <body>
    <header>
      
    </header>

    <nav>
      <ul>
        <li><a href='businessHome.php' class='left'>Home</a></li>
        <li><a href='createJob.php' class='left'>Create Job</a></li>
        <li><a href='viewListings.php' class='left'>View Listings</a></li>

        <li class='dropdown right'>
          <a href='javascript:void(0)' class='dropdownBtn'>Profile ></a>
          <div class='dropdown-content'>
            <a href='profile.php'>View Profile</a>
            <a id='dropdownBottom' href=''>
              <form method='POST'>
                <input type='submit' id='logOutButton' name='logOutBtn' value='Log Out'>
              </form>
            </a>
          </div>
        </li>
      </ul>

      <?php
        if (isset($_POST['logOutBtn'])) {
          // Log out button has been pressed, destroy session and navigate to landing page
          session_destroy();
          header("Location:index.html");
        }
      ?>
    </nav>

    <main>
      <?php
        if (!isset($_SESSION['accountID'])) {
          // User has not logged in, navigate to userLogin.php
          header("Location:userLogin.php");
        }

        // Check if the account type is personal
        if ($_SESSION['accountType'] == "Personal") {

```

```

        // Account type is personal, navigate to userHome.php
        header("Location:userHome.php");
    }

    // Set error reporting to exclude warnings
    // Only fatal errors and parse errors are displayed
    error_reporting(E_ERROR | E_PARSE);

    // Database connection variables are established
    $dbHost = "localhost";
    $dbUser = "root";
    $dbPassword = "";
    $dbName = "project";

    // Attempt to connect to database
    $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

    // Check if the connection is successful, else display appropriate error message
    if (mysqli_connect_errno()) {
        echo "<h2>Connection Error</h2>";
        // Display connection error message
        echo mysqli_connect_error();
        die();
    }

    // Assign business ID to a variable
    $businessID = $_SESSION['accountID'];

    // Create query to select the 3 most recent job listings for the business
    $query = "SELECT * FROM job WHERE businessID = '$businessID' ORDER BY jobID DESC
LIMIT 3";

    // Execute the query using the active database
    $result = mysqli_query($connection, $query);

    // Check whether query has been successful
    if ($result) {
        // Count number of rows, if 1 or more are returned display most recent jobs
        $row_count = mysqli_num_rows($result);

        if ($row_count > 0) {
            // Create section for recent jobs
            echo "<section class='content'>
                <h3>Your Recent Listings</h3>";

            // Display most recent jobs in a section
            while ($row = mysqli_fetch_assoc($result)) {
                // Declare variables from database
                $jobID = $row['jobID'];
                $jobTitle = $row['title'];
                $briefDesc = htmlspecialchars_decode(substr($row['description'], 0, 100) . "...");
            }
        }
    }

```

```

        echo "<section class='flex card'>
            <section class='flexChild75'>
                <h3>${jobTitle}</h3>
                <p>${briefDesc}</p>
            </section>
            <section class='centerText flexCenter flexChild25'>
                <a href='viewApplicants.php?job=${jobID}'><button type='button'>View
Applicants</button></a>
            </section>
        </section>";
    }

    // Close recent jobs section
    echo "</section>";
}

else {
    // If no records are returned then no jobs were found
    echo "<section class='content'>
        <h3>No recent job listings</h3>
        <p>You haven't made any job listings yet, try creating your first one <a
href='createJob.php'>here</a>.</p>
    </section>";
}

}

else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

// Close the database connection
mysqli_close($connection);
?>
</main>
</body>
</html>

```

3.2.11 Create New Job Page

createJob.php

```

<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Create a Job</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

```

```

</head>

<body>
  <header>
    
  </header>

  <nav>
    <ul>
      <li><a href='businessHome.php' class='left'>Home</a></li>
      <li><a href='createJob.php' class='left'>Create Job</a></li>
      <li><a href='viewListings.php' class='left'>View Listings</a></li>

      <li class='dropdown right'>
        <a href='javascript:void(0)' class='dropdownBtn'>Profile ></a>
        <div class='dropdown-content'>
          <a href='profile.php'>View Profile</a>
          <a id='dropdownBottom' href=''>
            <form method='POST'>
              <input type='submit' id='logOutButton' name='logOutBtn' value='Log Out'>
            </form>
          </a>
        </div>
      </li>
    </ul>

    <?php
      if (isset($_POST['logOutBtn'])) {
        // Log out button has been pressed, destroy session and navigate to landing page
        session_destroy();
        header("Location:index.html");
      }
    ?>
  </nav>

  <main>
    <?php
      if (!isset($_SESSION['accountID'])) {
        // User has not logged in, navigate to userLogin.php
        header("Location:userLogin.php");
      }

      // Check if the account type is personal
      if ($_SESSION['accountType'] == "Personal") {
        // Account type is personal, navigate to userHome.php
        header("Location:userHome.php");
      }

      // Check if form has been submitted
      if (isset($_POST['createJob'])) {
        // Set error reporting to exclude warnings

```

```

// Only fatal errors and parse errors are displayed
error_reporting(E_ERROR | E_PARSE);

// Database connection variables are established
$dbHost = "localhost";
$dbUser = "root";
$dbPassword = "";
$dbName = "project";

// Attempt to connect to database
$connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

// Check if the connection is successful, else display appropriate error message
if (mysqli_connect_errno()) {
    echo "<h2>Connection Error</h2>";
    // Display connection error message
    echo mysqli_connect_error();
    die();
}

// Create variable for businessID
$businessID = $_SESSION['accountID'];

// Store data from html form in php variables
$jobTitle = $_POST['jobTitle'];
$jobType = $_POST['jobType'];
$hourlyRate = $_POST['hourlyRate'];
$weeklyHours = $_POST['weeklyHours'];
$jobLocation = $_POST['jobLocation'];
$description = htmlspecialchars($_POST['description']);

// Create query to insert job into database
$query = "INSERT INTO job(title, jobType, businessID, description, weeklyHours,
hourlyRate, location) VALUES('$jobTitle', '$jobType', '$businessID', '$description', '$weeklyHours',
'$hourlyRate', '$jobLocation')";

// Execute the query using the active database
$result = mysqli_query($connection, $query);

// Check whether query was successful
if ($result) {
    // Query was successful and application has been made
    echo "<section class='content'>
        <h2>Job Created</h2>
        <p>You have created the position $jobTitle!</p>
        <p>View all of your listings <a href='listings.php'>here</a>.</p>
    </section>";
}
else {
    // If query was unsuccessful display error message
    echo "<p>Error creating job</p>";
}

```

```

        echo "<p>Error description: " . mysqli_error($connection) . "</p>";
    }

    // Close the database connection
    mysqli_close($connection);
}
?>

<section class="content">
    <h2>Create a Job</h2>

    <form action="createJob.php" method="POST">
        <section class="flex">
            <section id="jobDetails" class="flexChild50" style="margin-right: 30px;">
                <label for="title">Title</label><br>
                <input type="text" name="jobTitle" id="title" required><br><br>

                <label for="type">Job Type</label><br>
                <select name="jobType" id="type">
                    <option value="Part-time">Part-time</option>
                    <option value="Full-time">Full-time</option>
                    <option value="Apprenticeship">Apprenticeship</option>
                </select><br><br>

                <label for="hrRate">Hourly Rate</label><br>
                <input type="text" name="hourlyRate" id="hrRate" required><br><br>

                <label for="wkHours">Weekly Hours</label><br>
                <input type="text" name="weeklyHours" id="wkHours" required><br><br>

                <label for="location">Location</label><br>
                <input type="text" name="jobLocation" id="location" required><br><br>
            </section>

            <section class="flexChild50">
                <label for="descriptionTextArea">Description</label>
                <textarea name="description" id="descriptionTextArea" maxlength="4000"
required></textarea>
                <p id="descriptionCharsRemaining">4000 Characters Remaining</p>

                <section id="createJob">
                    <input type="submit" name="createJob" value="Create" id="createJobBtn">
                </section>
            </section>
        </section>
    </form>
</section>
</main>

<script>
    const textarea = document.getElementById("descriptionTextArea");

```

```

        textarea.addEventListener("input", event => {
            const target = event.currentTarget;
            const maxLength = target.getAttribute("maxlength");
            const currentLength = target.value.length;
            const charsRemaining = maxLength - currentLength;

            document.getElementById("descriptionCharsRemaining").innerText = charsRemaining + "
Characters Remaining";
        });
    </script>
</body>
</html>

```

3.2.12 View Job Listings Page

viewListings.php

```

<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>Listings</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <header>
            
        </header>

        <nav>
            <ul>
                <li><a href='businessHome.php' class='left'>Home</a></li>
                <li><a href='createJob.php' class='left'>Create Job</a></li>
                <li><a href='viewListings.php' class='left'>View Listings</a></li>

                <li class='dropdown right'>
                    <a href='javascript:void(0)' class='dropdownBtn'>Profile ></a>
                    <div class='dropdown-content'>
                        <a href='profile.php'>View Profile</a>
                        <a id='dropdownBottom' href=''>
                            <form method='POST'>
                                <input type='submit' id='logOutButton' name='logOutBtn' value='Log Out'>
                            </form>
                        </a>
                    </div>
                </li>
            </ul>
        </nav>
    </body>
</html>

```

```

        </li>
    </ul>

    <?php
        if (isset($_POST['logOutBtn'])) {
            // Log out button has been pressed, destroy session and navigate to landing page
            session_destroy();
            header("Location:index.html");
        }
    ?>
</nav>

<main>
    <?php
        if (!isset($_SESSION['accountID'])) {
            // User has not logged in, navigate to userLogin.php
            header("Location:userLogin.php");
        }

        // Check if the account type is personal
        if ($_SESSION['accountType'] == "Personal") {
            // Account type is personal, navigate to userHome.php
            header("Location:userHome.php");
        }

        // Set error reporting to exclude warnings
        // Only fatal errors and parse errors are displayed
        error_reporting(E_ERROR | E_PARSE);

        // Database connection variables are established
        $dbHost = "localhost";
        $dbUser = "root";
        $dbPassword = "";
        $dbName = "project";

        // Attempt to connect to database
        $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

        // Check if the connection is successful, else display appropriate error message
        if (mysqli_connect_errno()) {
            echo "<h2>Connection Error</h2>";
            // Display connection error message
            echo mysqli_connect_error();
            die();
        }

        // Assign business ID to a variable
        $businessID = $_SESSION['accountID'];

        // Create query to select all job listings for the business
        $query = "SELECT * FROM job WHERE businessID = '$businessID' ORDER BY jobID DESC";
    
```

```

// Execute the query using the active database
$result = mysqli_query($connection, $query);

// Check whether query has been succesful
if ($result) {
    // Count number of rows, if 1 or more are returned display most recent jobs
    $row_count = mysqli_num_rows($result);

    if ($row_count > 0) {
        // Create section for recent jobs
        echo "<section class='content'>
            <h3>Your Listings</h3>";

        // Display most recent jobs in a section
        while ($row = mysqli_fetch_assoc($result)) {
            // Declare variables from database
            $jobID = $row['jobID'];
            $jobTitle = $row['title'];
            $briefDesc = htmlspecialchars_decode(substr($row['description'], 0, 100) . "...");

            echo "<section class='flex card'>
                <section class='flexChild75'>
                    <h3>$jobTitle</h3>
                    <p>$briefDesc</p>
                </section>
                <section class='centerText flexCenter flexChild25'>
                    <a href='viewApplicants.php?job=$jobID'><button type='button'>View
Applicants</button></a>
                </section>
            </section>";
        }

        // Close recent jobs section
        echo "</section>";
    }
    else {
        // If no records are returned then no jobs were found
        echo "<section class='content'>
            <h3>No job listings</h3>
            <p>You haven't made any job listings yet, try creating your first one <a
href='createJob.php'>here</a>.</p>
        </section>";
    }
}
else {
    // If results couldn't be returned display appropriate error message
    echo "<p>Error searching database</p>";
    echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

```

```

        // Close the database connection
        mysqli_close($connection);
    ?>
</main>
</body>
</html>

```

3.2.13 View Applicants Page

viewApplicants.php

```

<?php
    // Start session
    session_start();
?>
<!DOCTYPE html>
<html>
    <head>
        <title>View Applicants</title>
        <link rel="stylesheet" type="text/css" href="styles.css">
        <link rel="icon" type="image/x-icon" href="images/favicon.png">
    </head>

    <body>
        <header>
            
        </header>

        <nav>
            <ul>
                <li><a href='businessHome.php' class='left'>Home</a></li>
                <li><a href='createJob.php' class='left'>Create Job</a></li>
                <li><a href='viewListings.php' class='left'>View Listings</a></li>

                <li class='dropdown right'>
                    <a href='javascript:void(0)' class='dropdownBtn'>Profile ></a>
                    <div class='dropdown-content'>
                        <a href='profile.php'>View Profile</a>
                        <a id='dropdownBottom' href=''>
                            <form method='POST'>
                                <input type='submit' id='logOutButton' name='logOutBtn' value='Log Out'>
                            </form>
                        </a>
                    </div>
                </li>
            </ul>

            <?php
                if (isset($_POST['logOutBtn'])) {
                    // Log out button has been pressed, destroy session and navigate to landing page
                    session_destroy();
                }
            ?>

```

```

        header("Location:index.html");
    }
    ?>
</nav>

<main>
    <?php
        if (!isset($_SESSION['accountID'])) {
            // User has not logged in, navigate to userLogin.php
            header("Location:userLogin.php");
        }

        // Check if the account type is personal
        if ($_SESSION['accountType'] == "Personal") {
            // Account type is personal, navigate to userHome.php
            header("Location:userHome.php");
        }

        // Set error reporting to exclude warnings
        // Only fatal errors and parse errors are displayed
        error_reporting(E_ERROR | E_PARSE);

        // Database connection variables are established
        $dbHost = "localhost";
        $dbUser = "root";
        $dbPassword = "";
        $dbName = "project";

        // Attempt to connect to database
        $connection = mysqli_connect($dbHost, $dbUser, $dbPassword, $dbName);

        // Check if the connection is successful, else display appropriate error message
        if (mysqli_connect_errno()) {
            echo "<h2>Connection Error</h2>";
            // Display connection error message
            echo mysqli_connect_error();
            die();
        }

        // Assign data to PHP variables
        $jobID = $_GET['job'];
        $businessID = $_SESSION['accountID'];

        // Create query to display job details
        $query = "SELECT * FROM job WHERE job.jobID = '$jobID'";

        // Execute the query using the active database
        $result = mysqli_query($connection, $query);

        // Check whether query has been successful
        if ($result) {

```

```

// Store job details in accosciative array
$row = mysqli_fetch_assoc($result);

// Job details variables
$jobTitle = $row['title'];
$jobType = $row['jobType'];
$jobDesc = htmlspecialchars_decode(str_replace("\n", "<br>", $row['description']));
$hourlyRate = $row['hourlyRate'];
$wkHours = $row['weeklyHours'];
$location = $row['location'];

echo "<section class='content'>
    <h2>$jobTitle</h2>
    <details>
        <summary>View job description</summary>
        <blockquote>$jobDesc</blockquote>
    </details>
    <br>
</section>";

// Open flex section
echo "<section class='flex' style='margin-top: -10px;'>";

echo "<section class='content flexChild25'>";

// Display job title
echo "<h3>Job details</h3>";

// Display job details
echo "<table class='jobDetails'>
    <tr>
        <td class='tdShade'>Job Type</td>
        <td>$jobType</td>
    </tr>
    <tr>
        <td class='tdShade'>Hourly Rate</td>
        <td>$hourlyRate</td>
    </tr>
    <tr>
        <td class='tdShade'>Weekly Hours</td>
        <td>$wkHours</td>
    </tr>
    <tr>
        <td class='tdShade'>Location</td>
        <td>$location</td>
    </tr>
</table>";

echo "</section>";
}
else {

```

```

// If results couldn't be returned display appropriate error message
echo "<p>Error searching database</p>";
echo "<p>Error description: " . mysqli_error($connection) . "</p>";
}

// Create query to display applicant details
$query = "SELECT * FROM user, application WHERE application.jobID = '$jobID' AND
user.userID = application.userID";

// Execute the query using the active database
$result = mysqli_query($connection, $query);

// Check whether query has been successful
if ($result) {
    // Count number of rows, if 1 or more are returned display applicants
    $row_count = mysqli_num_rows($result);

    if ($row_count > 0) {
        // Create section for recent jobs
        echo "<section class='content flexChild75'>
            <h3>Applicants</h3>";

        // Display applicants in a section
        while ($row = mysqli_fetch_assoc($result)) {
            // Declare variables from database
            $userName = $row['userName'];
            $userEmail = $row['email'];
            $coverLetter = htmlspecialchars_decode($row['coverLetter']);

            echo "<section class='card'>
                <h3>$userName</h3>
                <p>$userEmail</p>
                <details>
                    <summary>View cover letter</summary>
                    <blockquote>$coverLetter</blockquote>
                </details>
                <br>
            </section>";
        }

        // Close flex section
        echo "</section>";
    }
    else {
        // If no records are returned then no jobs were found
        echo "<section class='content flexChild75'>
            <h3>No Applicants Yet</h3>
            <p>No one has applied to this job yet but as soon as they do, they will show
up here!</p>
            </section>";
    }
}

```

```

    }
    else {
        // If results couldn't be returned display appropriate error message
        echo "<p>Error searching database</p>";
        echo "<p>Error description: " . mysqli_error($connection) . "</p>";
    }

    // Close the database connection
    mysqli_close($connection);
?>
</main>
</body>
</html>

```

3.2.14 CSS Styles

```

styles.css
/* Import font */
@import url('https://fonts.googleapis.com/css2?family=Poppins&display=swap');

/* Set global styles such as font */
* {
    font-family: "Poppins", sans-serif;
}
b {
    color: rgb(200, 47, 80);
}

body {
    margin: 0;
}
header {
    margin: 0;
    overflow: hidden;
    background-color: rgb(234, 234, 234);
}
header img {
    height: 100px;
}

.content {
    padding: 20px 30px 20px 30px;
    background-color: rgb(244, 244, 244);
    margin: 10px;
    margin-top: 20px;
    border-radius: 16px;
    box-shadow: 0 10px 10px -5px rgb(0 0 0 / 0.2);
}
.card {
    padding: 10px 15px 10px 15px;
}

```

```

    background-color: rgb(250,250,250);
    margin: 5px;
    margin-top: 10px;
    border-radius: 16px;
    box-shadow: 0 10px 10px -5px rgb(0 0 0 / 0.2);
    border: 1px solid black;
}
blockquote {
    border-left: 10px solid rgb(214, 214, 214);
    padding-left: 10px;
    margin-left: 0;
}

button {
    padding: 5px;
    border-radius: 10px;
    border: 1px solid black;
}
button:hover, input[type="submit"]:hover {
    background-color: darkgrey;
}

.links button {
    width: 150px;
}

.centerBox {
    position: absolute;
    top: 50%;
    left: 50%;
    transform: translate(-50%, -50%);
    border: 1px solid black;
    border-radius: 16px;
}

.centerText {
    text-align: center;
}

.loginForm {
    padding: 20px 30px 20px 30px;
}
#loginBox {
    width: 400px;
    box-shadow: 0 10px 10px -5px rgb(0 0 0 / 0.2);
}
#loginBox input[type="text"], #loginBox input[type="email"], #loginBox input[type="password"] {
    width: 100%;
    border-radius: 6px;
    border: 1px solid black;
}
input[type="submit"] {

```

```

    border-radius: 6px;
    border: 1px solid black;
}

#jobSearchForm input[type="text"], #jobSearchForm select, #jobDetails input[type="text"],
#jobDetails select {
    width: 100%;
    border-radius: 5px;
    border: 1px solid black;
}

td {
    padding-left: 20px;
}

.jobDetails {
    width: 100%;
    border: 1px solid black;
    border-collapse: collapse;
}

.jobDetails td {
    border: 1px solid black;
}

.jobDetails .tdShade {
    background-color: rgb(224, 224, 224);
}

textarea {
    width: 100%;
    height: 100px;
    border-radius: 5px;
}

#descriptionTextArea {
    height: 200px;
}

#coverCharsRemaining, #descriptionCharsRemaining {
    text-align: right;
    color: gray;
    font-style: italic;
}

#submitApplication, #createJob {
    width: 100%;
    text-align: right;
}

#submitBtn, #createJobBtn {
    width: 150px;
    padding: 10px;
    margin-top: 10px;
    margin-right: 5px;
}

.flex {
    display: flex;

```

```

}
.flexChild25 {
  width: 25%;
}
.flexChild50 {
  width: 50%;
}
.flexChild75 {
  width: 75%;
}
.flexCenter {
  display: flex;
  justify-content: center;
  align-items: center;
}

.left {
  float: left;
}
.right {
  float: right;
}

nav {
  background-color: rgb(234, 234, 234);
  width: 100%;
  margin-top: -7px;
  box-shadow: 0 10px 10px -5px rgb(0 0 0 / 0.2);
}
nav ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  overflow: hidden;
}
nav ul li {
  text-align: center;
}
nav ul li a {
  display: block;
  padding: 10px;
  text-decoration: none;
  color: black;
}
nav ul li a:hover {
  background-color: grey;
  color: white;
}
li.dropdown {
  display: inline-block;
}

```

```

nav ul.dropdown-content {
  display: none;
  position: absolute;
  /* background-color: #f9f9f9; */
  background-color: rgb(234, 234, 234);
  min-width: 160px;
  right: 0px;
  box-shadow: 0 10px 10px -5px rgb(0 0 0 / 0.2);
  z-index: 1;
  border-bottom-left-radius: 16px;
  border-bottom-right-radius: 16px;
}
nav ul.dropdown-content a {
  color: black;
  padding: 12px 16px;
  text-decoration: none;
  display: block;
  text-align: left;
}
nav ul.dropdown:hover .dropdown-content {
  display: block;
}
#dropdownBottom {
  border-bottom-left-radius: 16px;
  border-bottom-right-radius: 16px;
}
#logOutButton {
  background: none;
  border: none;
  text-align: left;
  padding: 0;
  width: 100%;
  height: 100%;
  font-size: 16px;
}
#logOutButton:hover {
  background-color: grey;
  color: white;
  cursor: pointer;
}

#tabSelect {
  list-style-type: none;
  margin: 0;
  padding: 0;
  overflow: hidden;
}
#tabSelect li {
  float: left;
  width: 50%;
  text-align: center;
}

```

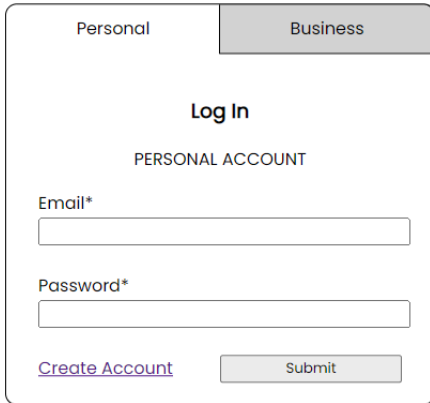
```

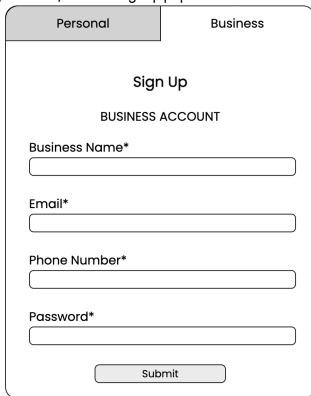
}
#tabSelect li a {
  display: block;
  padding: 10px;
  text-decoration: none;
  color: black;
}
#tabSelect li a:not(.active){
  background-color: lightgrey;
  outline: 1px solid black;
  border-bottom: 1px solid black;
}
#tabSelect li a:hover {
  background-color: grey;
}

@media screen and (max-width: 600px) {
  main {
    padding: 10px;
  }
  .flex {
    max-width: 100%;
    display: block;
  }
  .flexChild25, .flexChild50, .flexChild75 {
    width: 100%;
  }
  .content {
    padding: 20px 0 20px 0;
    margin: 10px 0 10px 0;
  }
  .card {
    margin: 20px;
  }
}

```

3.3 Ongoing Testing

3.3.1 Test Connection to Database	
Description of problem	When attempting to connect to the database, I mistyped the name of the database table which caused the website to throw a connection error. Website output: Error searching database Error description: Table 'project.users' doesn't exist  Original query: <pre>\$query = "SELECT * FROM users WHERE users.email = '\$email' AND users.password = '\$pword'";</pre>
Resolution	The name of the database table is 'user' and not 'users' so I have amended my query to use the correct table name. The website now connects to the database as expected and allows the user to log in. Amended query: <pre>\$query = "SELECT * FROM user WHERE user.email = '\$email' AND user.password = '\$pword'";</pre>

3.3.2 Test Creating a Business Account	
Description of problem	When creating a business account, there is a PHP fatal error as shown. Fatal error: Uncaught mysqli_sql_exception: Data truncated for column 'phoneNo' at row 1 in /Applications/MAMP/htdocs/PROJECT/businessSignUp.php:84 Stack trace: #0 /Applications/MAMP/htdocs/PROJECT/businessSignUp.php(84): mysqli_query(Object(mysqli), 'INSERT INTO bus...') #1 {main} thrown in /Applications/MAMP/htdocs/PROJECT/businessSignUp.php on line 84 

	This is caused by the phone number being stored as an integer in the database and the database attempting to truncate the value stored.										
Resolution	The database now stores phone numbers as a varchar instead of an integer and a business account can be created without throwing a fatal error.										
3.3.3 Testing the Apply for Job Page											
Description of problem	When searching the database for the job and business details, the wrong business was displayed on the details page. Apply for Device Expert <table border="1"> <tr> <td>Job Type</td><td>Part-time</td></tr> <tr> <td>Weekly Hours</td><td>12.00</td></tr> </table> Business Details <table> <tr> <td>Name</td><td>Vintage Clothes</td></tr> <tr> <td>Email</td><td>contact@vintageclothes.com</td></tr> <tr> <td>Phone Number</td><td>01632 363826</td></tr> </table> For example, the business here should've been <i>Horizon Tech Innovations</i> however was incorrectly displaying as <i>Vintage Clothes</i>. <code>\$query = "SELECT * FROM job, business WHERE job.jobID = '\$jobID' and job.jobID = business.businessID";</code> This is because the query compares job.jobID = business.businessID instead of job.businessID = business.businessID	Job Type	Part-time	Weekly Hours	12.00	Name	Vintage Clothes	Email	contact@vintageclothes.com	Phone Number	01632 363826
Job Type	Part-time										
Weekly Hours	12.00										
Name	Vintage Clothes										
Email	contact@vintageclothes.com										
Phone Number	01632 363826										
Resolution	The query has been adjusted to correctly compare the businessID of the two tables and the correct business details is now showing for each job.										

3.3.4 Cover letter not inserting to database	
Description of problem	When attempting to apply for a job if certain characters were in the cover letter, such as quotations, it would fail to store in the database and an error message was displayed.  Home Job Search Applications Profile > Fatal error: Uncaught mysqli_sql_exception: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''')' at line 1 in /Applications/MAMP/htdocs/PROJECT/apply.php:130 Stack trace: #0 /Applications/MAMP/htdocs/PROJECT/apply.php(130): mysqli_query(Object(mysqli), 'INSERT INTO app...') #1 {main} thrown in /Applications/MAMP/htdocs/PROJECT/apply.php on line 130 The original code did not change any 'special' characters, such as single quotes, before attempting to store the data in the databases. The original code is: <code>\$coverLetter = \$_POST['coverLetter'];</code>

Resolution	To fix this, I used a PHP function htmlspecialchars which removed any special characters from the text before storing it in the database. <pre><code>\$coverLetter = htmlspecialchars(\$_POST['coverLetter']);</code></pre> When displaying data from the database I now need to call the PHP function htmlspecialchars_decode to get the special characters back.
------------	--

3.3.5 Create job doesn't set location							
Description of problem	When a business creates a job, all details are added to the database however the location remains blank. Job Created You have created the position Warehouse Operator! View all of your listings here. As shown the success message is displayed on the website but when I checked the database the data has not inserted correctly. <table><thead><tr><th>weeklyHours</th><th>hourlyRate</th><th>location</th></tr></thead><tbody><tr><td>40.00</td><td>12.00</td><td></td></tr></tbody></table> As shown the location field remains blank even once the form has been submitted. This is because I mistyped the PHP variable in the insert query. Original code: <pre>\$query = "INSERT INTO job(title, jobType, businessID, description, weeklyHours, hourlyRate, location) VALUES('\$jobTitle', '\$jobType', '\$businessID', '\$description', '\$weeklyHours', '\$hourlyRate', '\$location')";</pre> The last variable <i>\$location</i> should be <i>\$jobLocation</i>.	weeklyHours	hourlyRate	location	40.00	12.00	
weeklyHours	hourlyRate	location					
40.00	12.00						
Resolution	I have changed the variable to be typed correctly in the query and the data now inserts into the database as expected. New code: <pre>\$query = "INSERT INTO job(title, jobType, businessID, description, weeklyHours, hourlyRate, location) VALUES('\$jobTitle', '\$jobType', '\$businessID', '\$description', '\$weeklyHours', '\$hourlyRate', '\$jobLocation')";</pre>						

3.4 New Skills and Knowledge

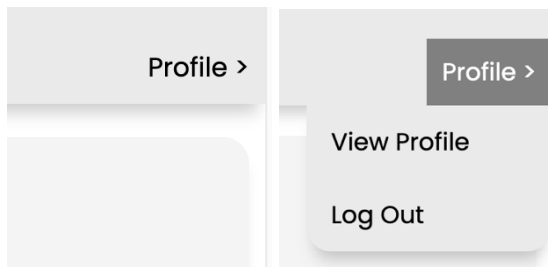
3.4.1 Count the Number of Characters in a Text Area	
The text areas on the apply for job page and create a job page have a character limit and I wanted to display how many characters the user had left.	

Source: Stack Overflow

<https://stackoverflow.com/questions/14086398/count-textarea-characters>

3.4.2 Dropdown Nav Bar Items

The profile button on the nav bar expands a dropdown menu when hovered over to reveal two options, 'View Profile' and 'Log out'.



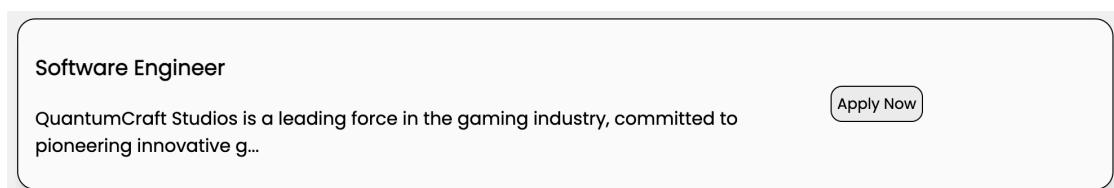
The image on the left is how the button displays normally and the image on the right when the cursor is hovered over the button.

Source: W3Schools

https://www.w3schools.com/css/css_navbar_horizontal.asp

3.4.3 PHP Substring

On the user and business home pages, job cards are displayed to show recent job listings relevant to the user or business. The full description would have been too long, so the PHP substring method was used to only display the first 100 characters of the description.



Only the first 100 characters of the description is displayed with an ellipsis.

Source: W3Schools

https://www.w3schools.com/php/func_string_substr.asp

3.4.4 HTML Special Characters

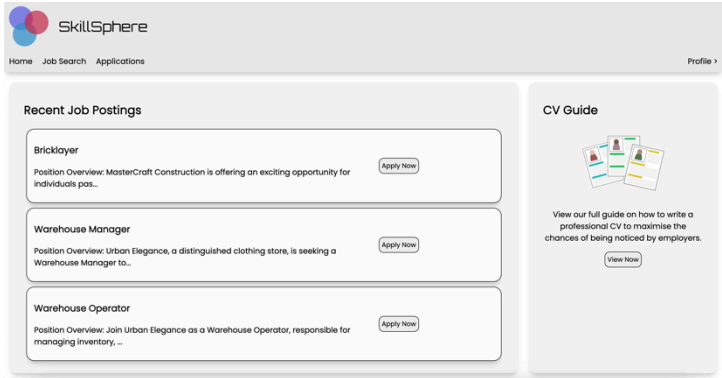
The text areas for cover letters and job descriptions would cause an error when inserting their content into the database if they contained HTML special characters such as a single quote. This involved using the `htmlspecialchars` and `htmlspecialchars_decode` to change the character into a database friendly character code.

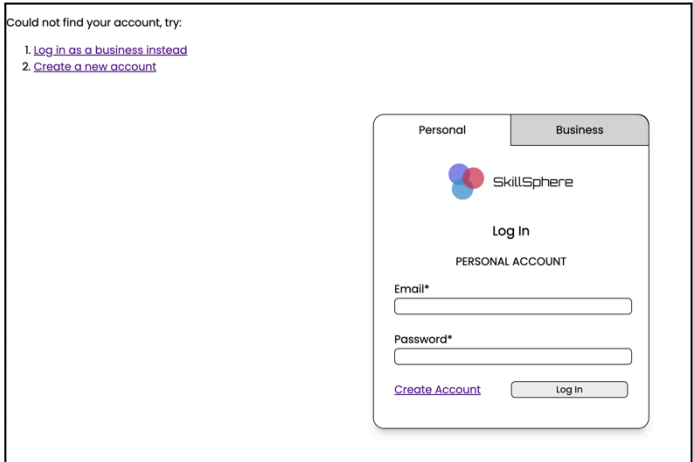
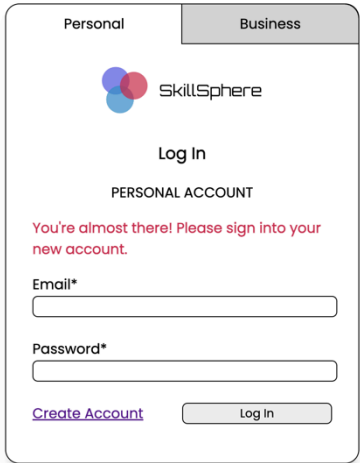
Source: Stack Overflow

<https://stackoverflow.com/questions/29112000/do-i-even-need-htmlspecialchars-for-text-areas-value>

4. Testing

4.1 Final Test Plan

Test ID	Test Description	Expected Result	Actual Result
1	Check that a user can log into an existing account. Case 1 – The user enters correct log in details. Case 2 – The user enters incorrect log in details.	Case 1 – The details are correct, and a row is returned from the database, the website accepts the details entered and navigates the user to <i>userHome.php</i> Case 2 – The details are incorrect, so no rows are returned from the database and the website rejects the details. An error message is displayed to the user.	Case 1 Results  The screenshot shows the SkillSphere website interface. At the top, there is a navigation bar with links for Home, Job Search, Applications, and a Profile link. Below the navigation bar, the main content area is divided into two sections. The left section, titled 'Recent Job Postings', lists three job opportunities: 'Bricklayer' (MasterCraft Construction), 'Warehouse Manager' (Urban Elegance), and 'Warehouse Operator' (Urban Elegance). Each listing includes a brief description and an 'Apply Now' button. The right section, titled 'CV Guide', features an icon of a CV and a link to 'View our full guide on how to write a professional CV to maximise the chances of being noticed by employers.' with a 'View Now' button.

			Case 2 Results Could not find your account, try: 1. Log in as a business instead 2. Create a new account 
2	Check that a user can create a new personal account. Case 1 – The user completes all fields with valid data. Case 2 – The user tries to create an account using an email that has already been registered. Case 3 – The user does not fill out all fields or enters data that does not meet the requirements (e.g. a password less than 8 characters).	Case 1 – The data is accepted and added to the database. The user is navigated to <i>userLogin.php</i> to sign in. Case 2 – An error message is displayed to say that the email is already in use. Case 3 – The form will not get submitted if the data does not meet the requirements (e.g. the password is too short, or a field is missing).	Case 1 Results 

Case 2 Results

This email is already in use, try [logging in](#) instead?

Personal

Business

 SkillSphere

Sign Up

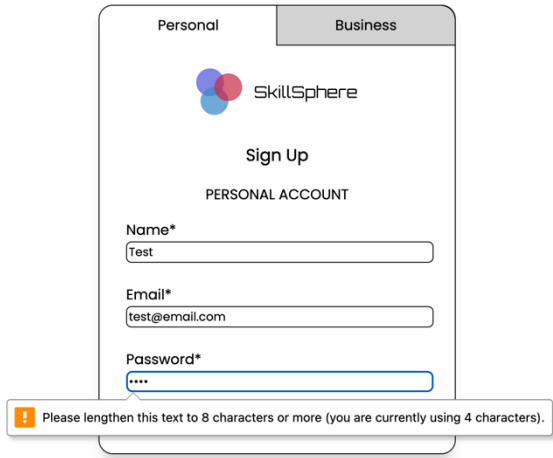
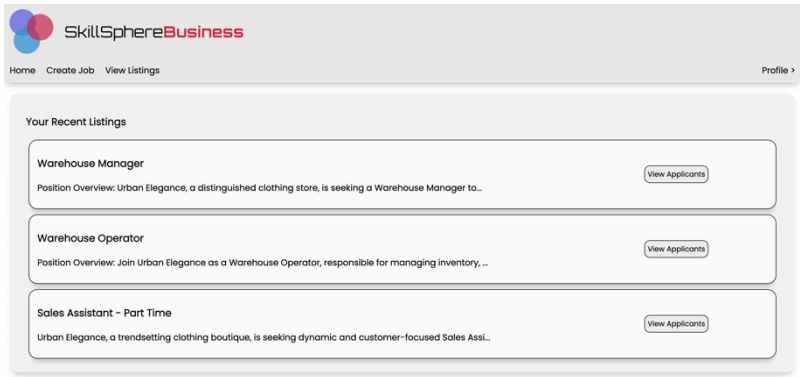
PERSONAL ACCOUNT

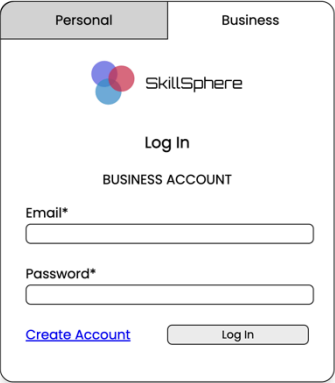
Name*

Email*

Password*

Create Account

			Case 3 Results 
3	Check that a business can log into an existing account. Case 1 – The business enters correct log in details. Case 2 – The business enters incorrect log in details.	Case 1 – The details are correct, and a row is returned from the database, the website accepts the details entered and navigates the business to <i>businessHome.php</i> Case 2 – The details are incorrect, so no rows are returned from the database and the website rejects the details. An error message is displayed to the business.	Case 1 Results 

			Case 2 Results Could not find your account, try: 1. Log in as a user instead 2. Create a new account
4	Check that a business can create a new business account. Case 1 – The business completes all fields with valid data. Case 2 – The business tries to create an account using an email that has already been registered. Case 3 – The business does not fill out all fields or enters data that does not meet the	Case 1 – The data is accepted and added to the database. The business is navigated to <i>businessLogin.php</i> to sign in. Case 2 – An error message is displayed to say that the email is already in use. Case 3 – The form will not get submitted if the data does not meet the requirements (e.g. the password is too short, or a field is missing).	

requirements (e.g. a password less than 8 characters).

Case 1 Results

Personal

Business

 SkillSphere

Log In

BUSINESS ACCOUNT

You're almost there! Please sign into your new account.

Email*

Password*

[Create Account](#)

Case 2 Results

This email is already in use, try [logging in](#) instead?

Personal

Business

 SkillSphere

Sign Up

BUSINESS ACCOUNT

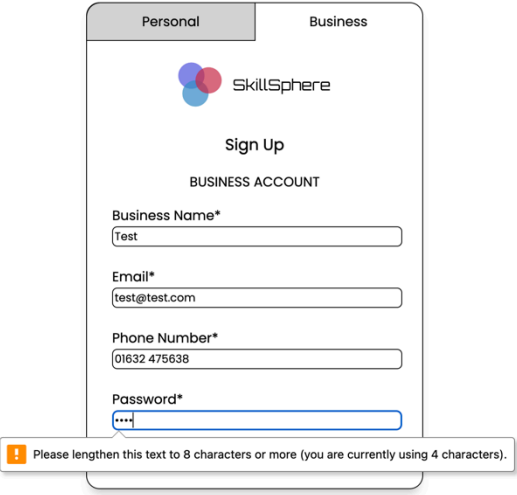
Business Name*

Email*

Phone Number*

Password*

Create Account

			Case 3 Results 
5	Check that the user can view details about a job and submit a job application. Case 1 – Viewing the details for the job. Case 2 – The user has submitted the application.	Case 1 – The user is presented with a table displaying key details about the job, the job description, an area to type a cover letter, and a button to submit the application. Case 2 – A message is displayed to the user saying that their application has been submitted.	

Case 1 Results

Apply for Bricklayer

Job Type	Apprenticeship	Hourly Rate	5.80
Weekly Hours	18.00	Location	Paisley

Business Details

Name MasterCraft Construction
Email construction@mastercraft.com
Phone Number 01632 962846

Job Description

Job description from database

Cover Letter (Recommended)

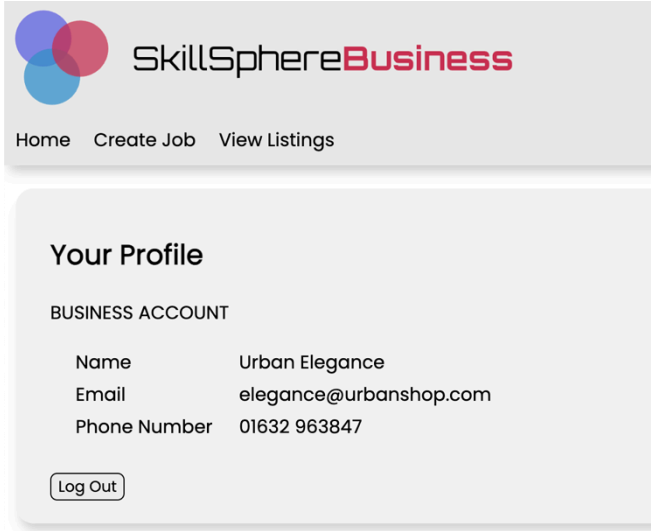
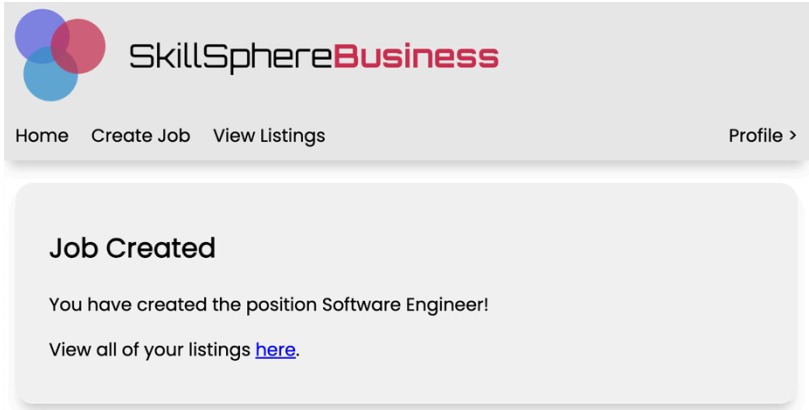
This is where you can tell the employer a little bit about yourself. Consider mentioning why you think you are ideal for the job.

2000 Characters Remaining

Submit

			Case 2 Results  Home Job Search Applications Application Successful You have applied for Device Expert at HorizonTech Innovations, good luck! View all of your applications here.
6	Check that all account types, user and business, can view account information on the profile page. Case 1 – The account type is 'Personal'. Case 2 – The account type is 'Business'.	Case 1 – The user can view their name and email. Case 2 – The business can view their name, email, and phone number. In both cases, there will also be a 'Log Out' button which will destroy the session and navigate back to <i>index.html</i> when pressed.	

			Case 1 Results  SkillSphere Home Job Search Applications Your Profile PERSONAL ACCOUNT Name Sam Richardon Email samrich03@email.com Log Out
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			Case 2 Results 
7	Check that a business can create a new job listing. Case 1 – The details are correctly input and valid. Case 2 – The details inputted are not valid or are missing.	Case 1 – The form submits, the data is inserted to the database and a success message is displayed to the business. Case 2 – The form does not submit if the data is not valid (e.g. missing, or too long).	Case 1 Results 

			Case 2 Results  SkillSphereBusiness Home Create Job View Listings Profile > Create a Job Title Software Engineer Job Type Full-time Hourly Rate 14.00 Weekly Hours 40 Location Glasgow Description Please fill in this field. remaining Create
8	Check that a business can view all their existing job listings and view all the applicants for each of them. Case 1 – When viewing <i>viewListings.php</i> to view all their existing job listings.	Case 1 – The page displays a list of each job that the business has created and a button to navigate to <i>viewApplicants.php</i> to see all the applicants for the job. Case 2 – The page displays the job details and description along with a list of all the applicants and their cover letters.	

Case 2 – When viewing *viewApplicants.php* to view the applicants of the job.

Case 1 Results

 SkillSphereBusiness

Home Create Job View Listings Profile >

Your Listings

Software Engineer

QuantumCraft Studios is a leading force in the gaming industry, committed to pioneering innovative g...

View Applicants

Case 2 Results

 SkillSphereBusiness

Home Create Job View Listings Profile >

Sales Assistant - Part Time

▼ View job description

Job description from database

Job details

Job Type	Part-time
Hourly Rate	11.80
Weekly Hours	20.00
Location	Renfrew

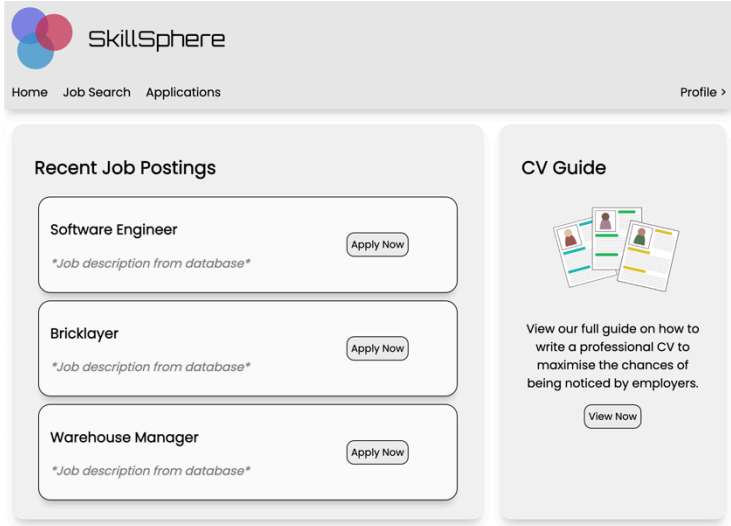
Applicants

Sam Richardson

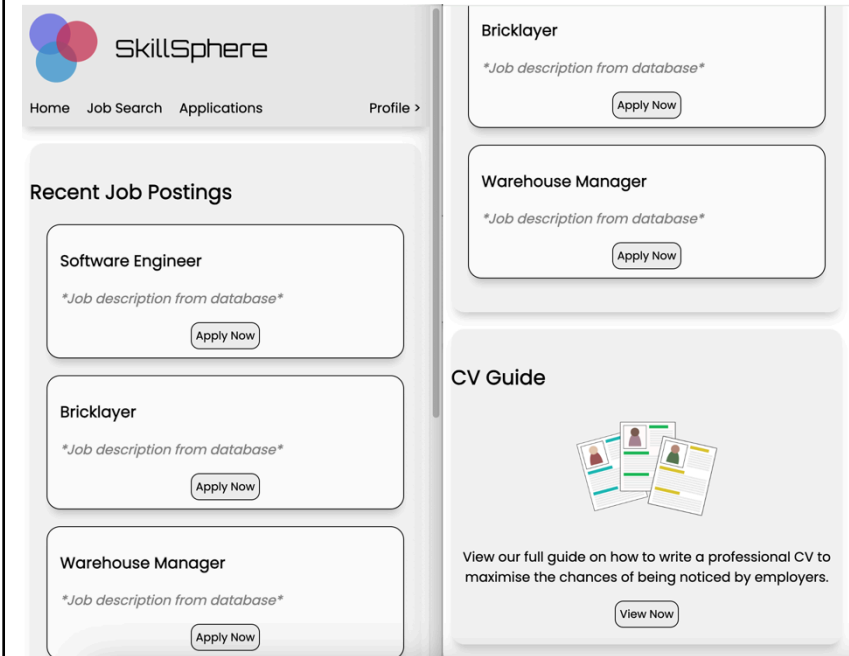
samrich03@email.com

▼ View cover letter

Cover letter from database

9	Check that the media query works and changes the flex sections to display as columns when the screen width is less than 600px. Case 1 – The screen width is greater than 600px. Case 2 – The screen width is less than 600px.	This is being demonstrated on <i>userHome.php</i> but applies to all pages. Case 1 – The flex sections are displayed normally in rows. Case 2 – The flex sections are displayed as columns.	Case 1 Results 
---	--	--	--

Case 2 Results



This image shows the top of the page (on the left) and the bottom of the page (on the right) to show that the CV guide section has moved to the bottom of the page.

4.2 Test Cases

4.2.1 Persona 1

Information – Lilly is a 16-year-old high school student who enjoys playing games on her computer in her spare time. She considers herself to be a skilled computer user and can use new apps and websites with ease. As she has recently turned 16, she is looking for her first part time job to earn some of her own money so she can go out with her friends more and start saving for university.

Test Results – For Lilly’s test she used the website to look for part time jobs that had suitable hours to work around her school days. The following key things were observed:

- Lilly found the website easy to use and intuitive to navigate. She created an account with ease and quickly navigated to the job search page where she applied filters to only see part time jobs. She found the job search filters to be useful as she could quickly find jobs with a specific pay rate or jobs in a specific town or city.
- She liked the fact that the website allowed any business to post new job listings as it means that there are a wide range of jobs available to apply for. She also appreciated how much information was displayed about the business she was applying to right on the job description page.
- She did say that it would be better if she could withdraw an existing application from her profile if she decides she no longer wants to be a candidate for the job.

Overall, this persona test suggests that the website meets the needs and goals of Lilly as she was able to complete all the tasks she wanted to, however there is some room for improvement regarding managing previous applications she has made.

4.2.2 Persona 2

Information – Oliver is a 28-year-old full time retail manager who considers himself an avid computer user but can sometimes struggle with new and unfamiliar apps and websites. Out with work he typically only uses computers for emails, web browsing and streaming videos and movies. He is hoping to find a new job with better employee benefits as he is unhappy with his current job.

Test Results – Oliver used the website to look for a new full-time job. The following key things were observed:

- Oliver liked the simple interface of the website as it made it easy for him to navigate despite sometimes struggling to use unfamiliar websites.
- He liked how easy it was to apply to jobs as all he needed to do was type a cover letter and submit. This made it easy to apply to multiple jobs in a short period of time which is useful when he is looking to change jobs as soon as possible.
- One thing he said that could be improved on the website is that there is to offer more filters when searching for jobs. For example, instead of a minimum number of hours, a range of hours would have been more useful.

Overall, Oliver liked the website, and this persona test suggests that the website is meeting his needs and goals as he was able to easily apply for new jobs. However, it has been suggested that there is room for improvement regarding the filters on the job search page to make it easier to find a job that is suitable to you.

4.2.3 Persona 3

Information – William is 52 years old, and he is a store owner at a local retail outlet and previously found it difficult to advertise new job positions online as he has limited computer experience and found other services to be more complex than his business requires. He is hoping for an easier solution to advertise new job positions that is simpler and more in line with what his business needs.

Test Results – William used the website to create a business profile and advertise some new positions at his shop. The following key things were observed:

- William liked the simplistic design and navigation of the website and that he could easily create a new job position. He likes how quickly he could create multiple positions within a short amount of time.
- He liked that he could see all the applicants, their contact details, and their cover letter in one place as he found that it makes reviewing potential candidates an easier task.
- One thing he would like to see improved on the website is information on how data is handled. As he has limited computer experience, he does not understand how the data is stored and is concerned that some sensitive information, such as candidate contact details, are visible publicly.

Overall, William had a positive experience with the website and wants to continue using it for his business to advertise new positions. He believes there is some room for improvement in terms of information on how data is stored and displayed so that it is more obvious what data can be viewed by anyone and what data is visible only to him.

5. Evaluation

5.1 Fitness for Purpose

My website is fit for purpose as the results of testing show that it meets all the requirements specified in *1.3 Requirements Specification*, including both end user and functional requirements. The website allows a new user to register for an account, log into an account, view job positions, apply for jobs and view all their existing applications. Likewise, the website allows a business to register for a new account, log into an account, create a new job and view all their existing job positions. The website validates all user input before storing it in the database whether it is the cover letter that a user can write when applying for a job, the job description a business writes when creating a new job, or even small inputs like the weekly hours on a new job position must be less than the number of hours in a week (168). I have also implemented a media query which responds to the screen size and adapts the layout of the website accordingly, such as for desktop screens or tablets, as specified. Furthermore, my persona testing shows that users and businesses successfully completed all assigned tasks on the website.

5.2 Robustness and Maintainability

5.2.1 Robustness of Code

My code is robust as it makes use of data validation. User input fields all have maximum lengths and appropriate types to ensure they can be inserted into the database without causing an error. For longer text areas where punctuation is likely to be used, the PHP `htmlspecialchars` function was used to ensure that all characters, including special characters, could be inserted into the database. For example, when a user or business creates an account, the database is checked to ensure that their email has not already been used to prevent duplicate accounts being created which would prevent the log in page from working properly if 2 or more accounts were returned.

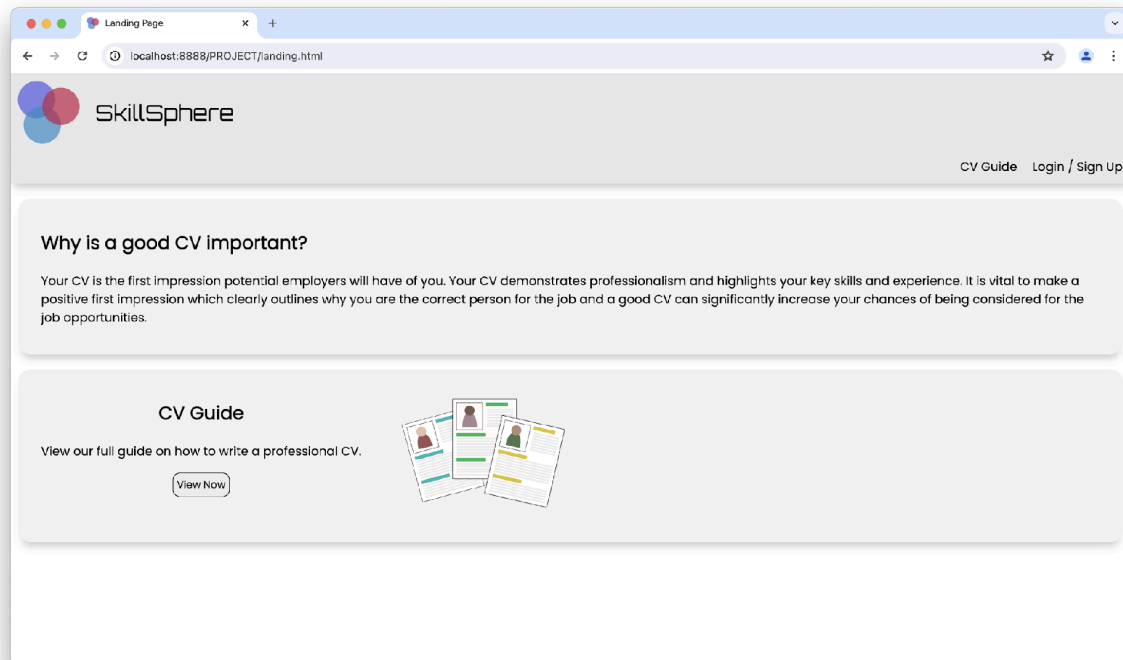
5.2.2 Maintainability of Code

My website is maintainable as I have used meaningful variable names in the HTML forms and PHP scripts which makes it clear as to what data each variable holds. Additionally clear, and descriptive comments have been added to each section of my PHP scripts to explain what the code is doing to aid readability. Moreover, good use of indentation and white spacing has been made to clearly group together related blocks of code to make it easier for other developers to read and understand the code. In addition, I have also made use of external styles which apply to every page which makes any form of maintenance in the future easier to carry out as each style only has to be changed once.

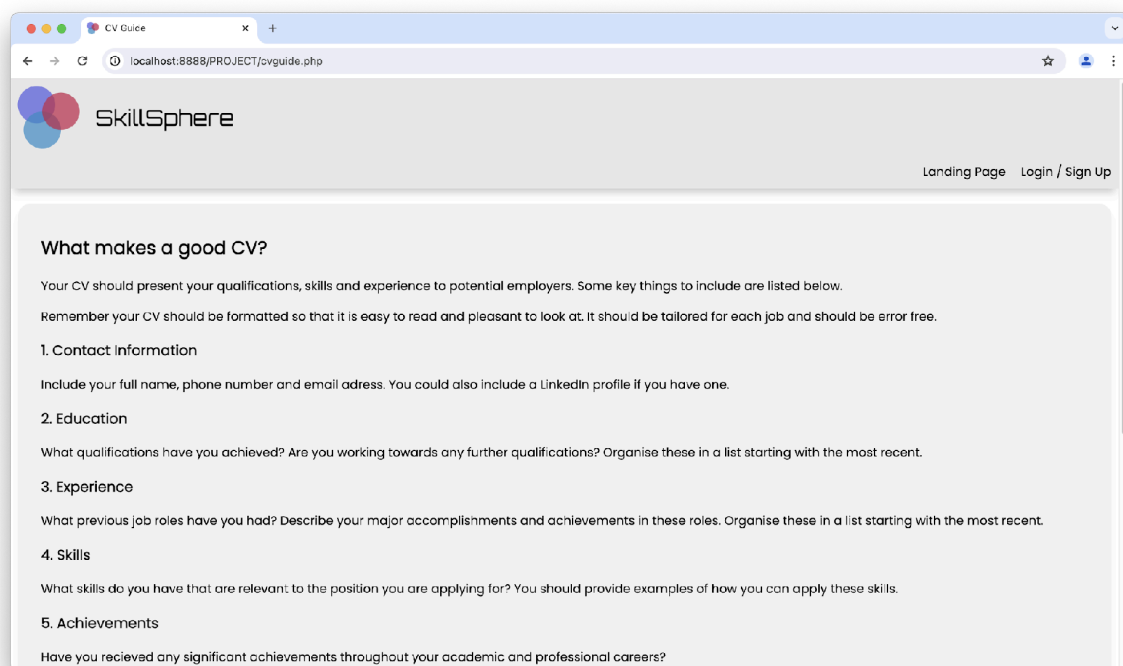
Appendix 1 – User Interface

Shown below are screenshots of all pages on the website to illustrate the user interface as viewed in a web browser (Google Chrome).

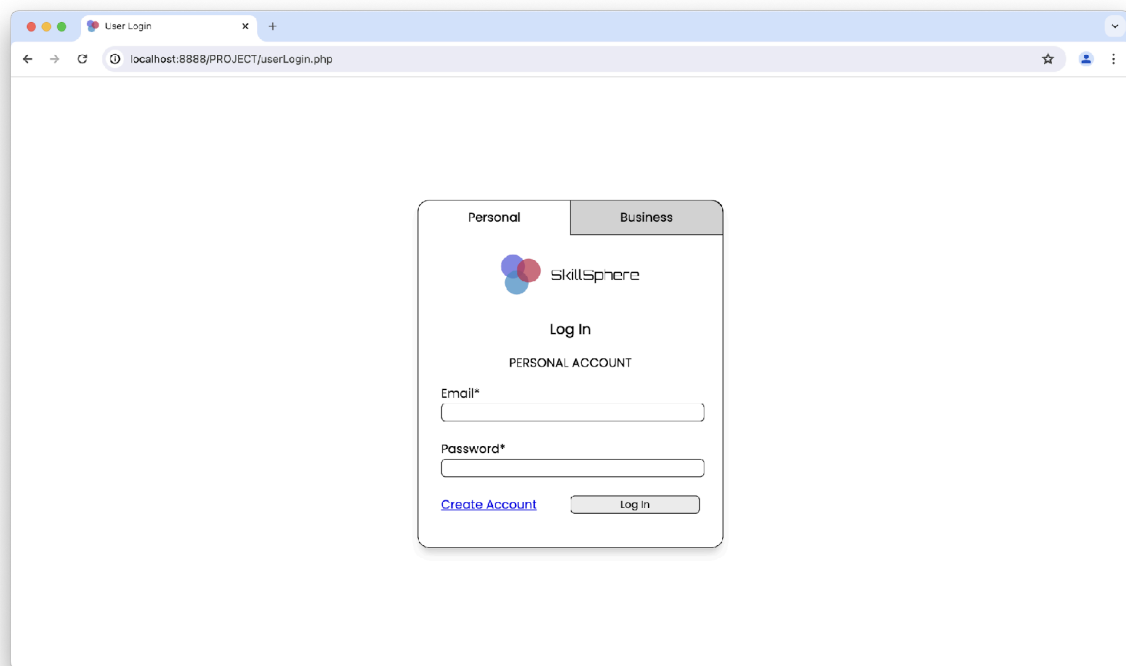
Landing Page



CV Guide Page



User Login Page



A screenshot of a web browser displaying the 'User Login' page. The browser's address bar shows 'localhost:8888/PROJECT/userLogin.php'. The page features a central login form with a 'Personal' tab selected and a 'Business' tab. The form includes the 'SkillsSphere' logo, the text 'Log In', and 'PERSONAL ACCOUNT'. It has input fields for 'Email*' and 'Password*', a 'Create Account' link, and a 'Log In' button.

Personal Business

 SkillsSphere

Log In

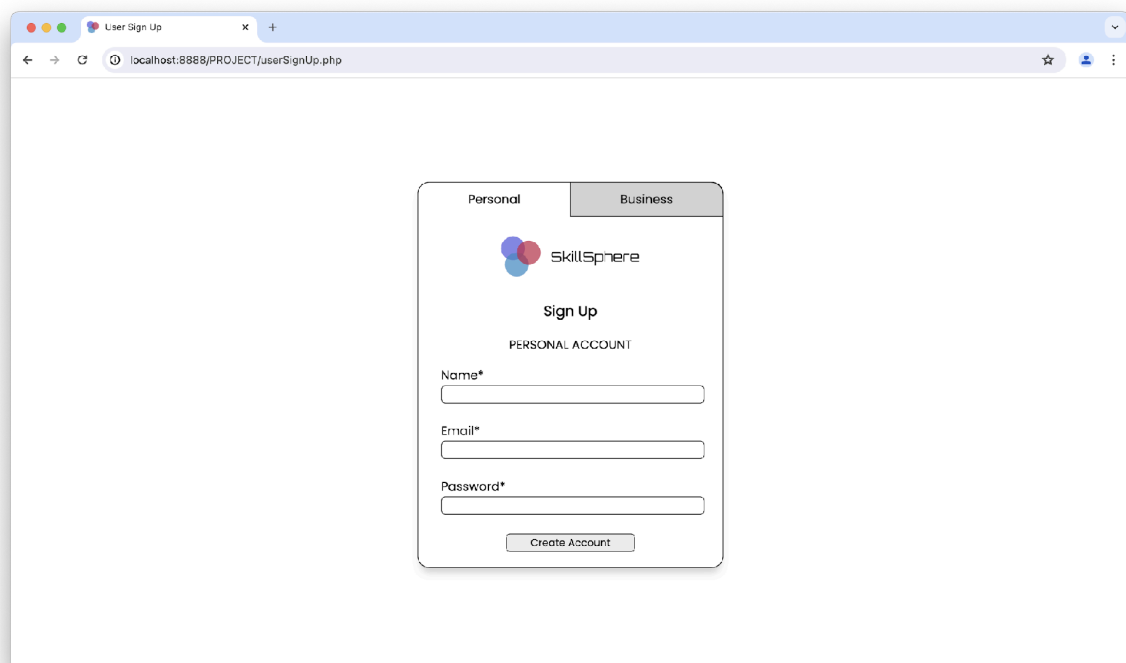
PERSONAL ACCOUNT

Email*

Password*

[Create Account](#)

User Sign Up Page



A screenshot of a web browser displaying the 'User Sign Up' page. The browser's address bar shows 'localhost:8888/PROJECT/userSignUp.php'. The page features a central sign-up form with a 'Personal' tab selected and a 'Business' tab. The form includes the 'SkillsSphere' logo, the text 'Sign Up', and 'PERSONAL ACCOUNT'. It has input fields for 'Name*', 'Email*', and 'Password*', and a 'Create Account' button.

Personal Business

 SkillsSphere

Sign Up

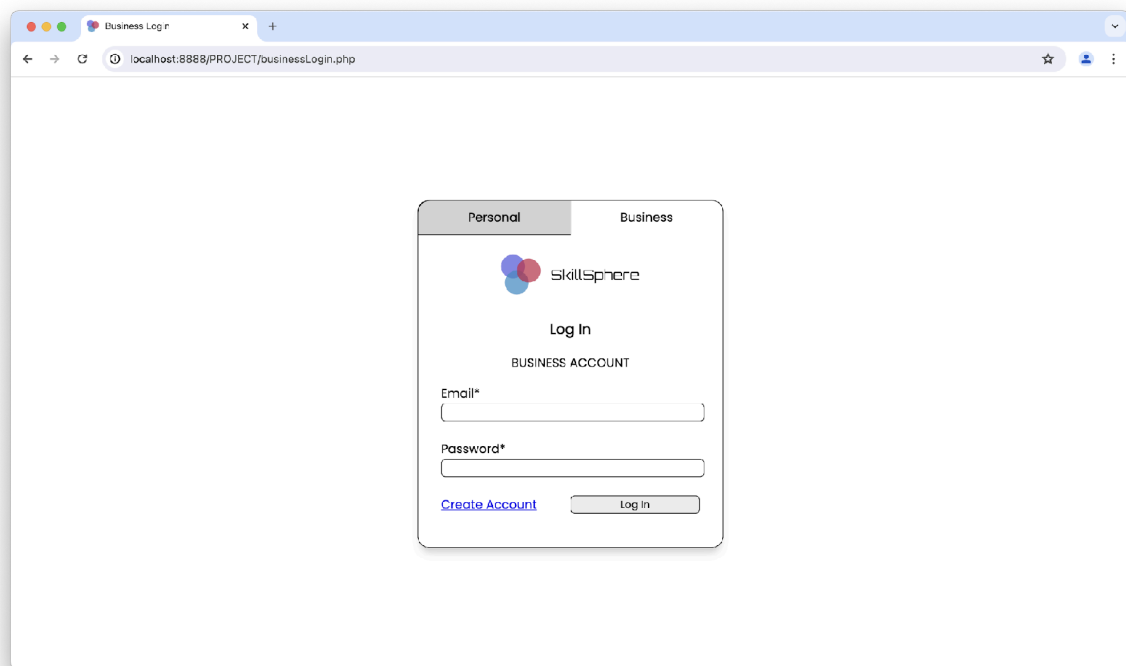
PERSONAL ACCOUNT

Name*

Email*

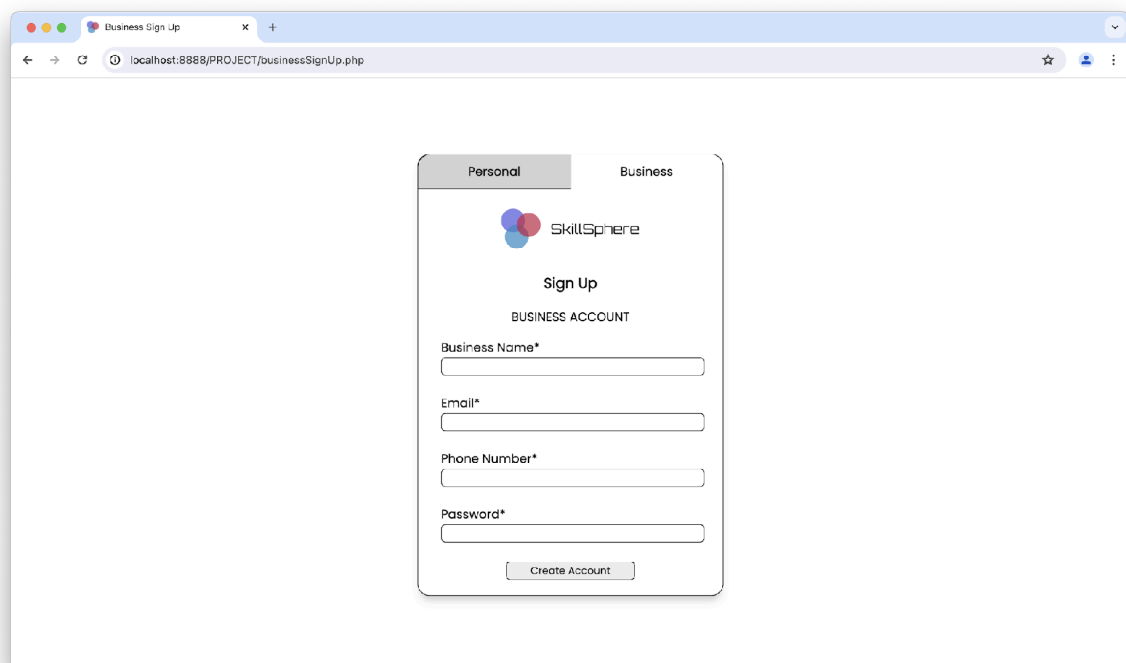
Password*

Business Login Page



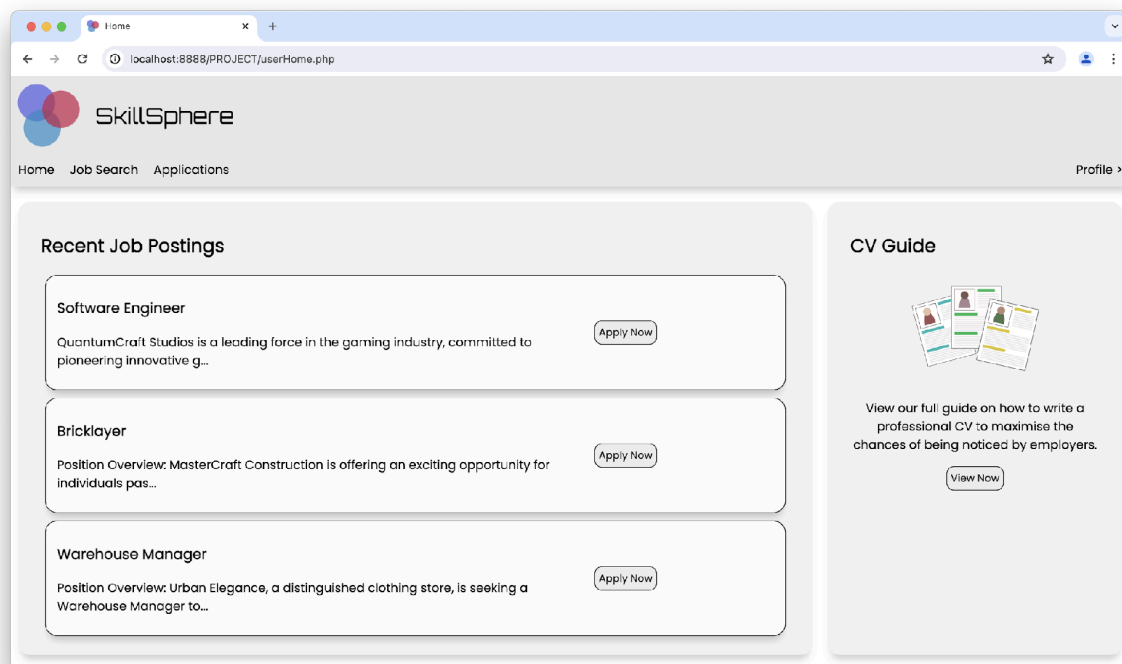
A screenshot of a web browser displaying the Business Login page. The browser's address bar shows the URL `localhost:8888/PROJECT/businessLogin.php`. The page features a central login form with a header section containing two tabs: "Personal" and "Business". The "Business" tab is selected. Below the tabs is the SkillSphere logo, which consists of three overlapping circles in blue, red, and green, followed by the text "SkillSphere". The form is titled "Log In" and "BUSINESS ACCOUNT". It includes two input fields: "Email*" and "Password*", each with a small asterisk indicating a required field. Below the password field is a blue link labeled "Create Account" and a grey button labeled "Log In".

Business Sign Up Page

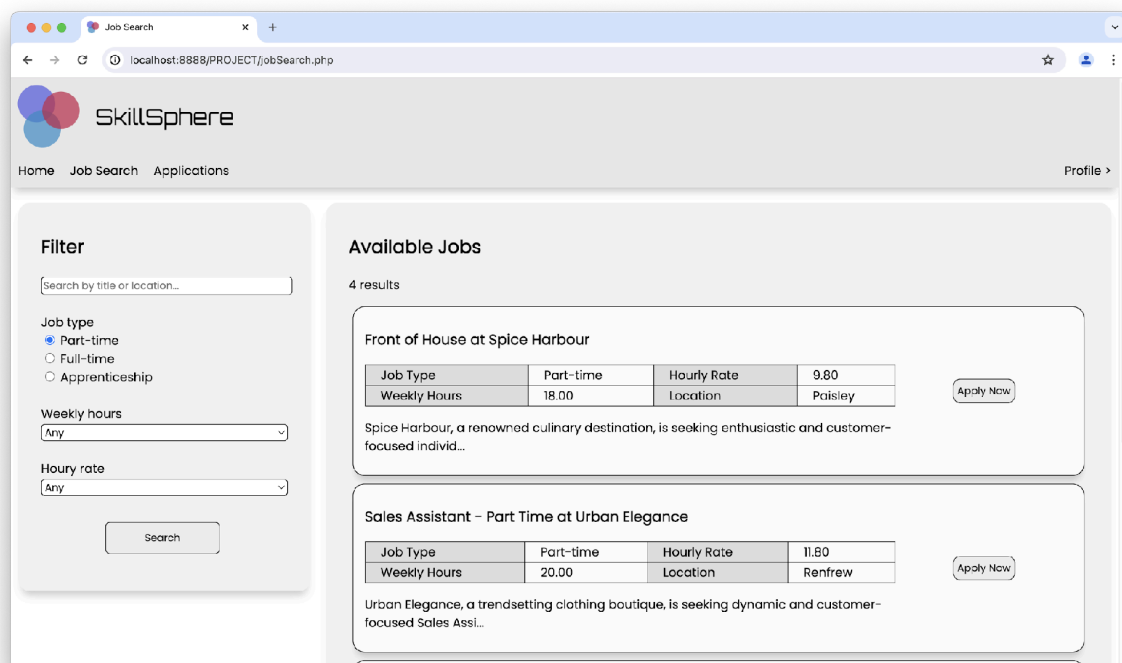


A screenshot of a web browser displaying the Business Sign Up page. The browser's address bar shows the URL `localhost:8888/PROJECT/businessSignUp.php`. The page features a central sign-up form with a header section containing two tabs: "Personal" and "Business". The "Business" tab is selected. Below the tabs is the SkillSphere logo, which consists of three overlapping circles in blue, red, and green, followed by the text "SkillSphere". The form is titled "Sign Up" and "BUSINESS ACCOUNT". It includes four input fields: "Business Name*", "Email*", "Phone Number*", and "Password*", each with a small asterisk indicating a required field. Below the password field is a grey button labeled "Create Account".

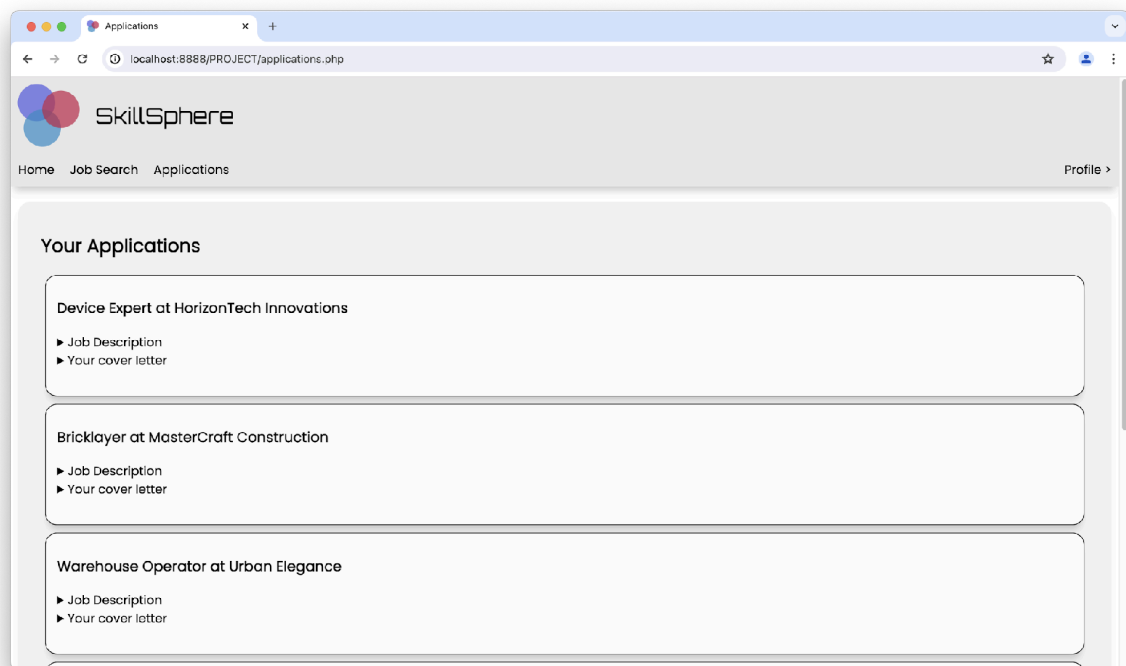
User Home Page



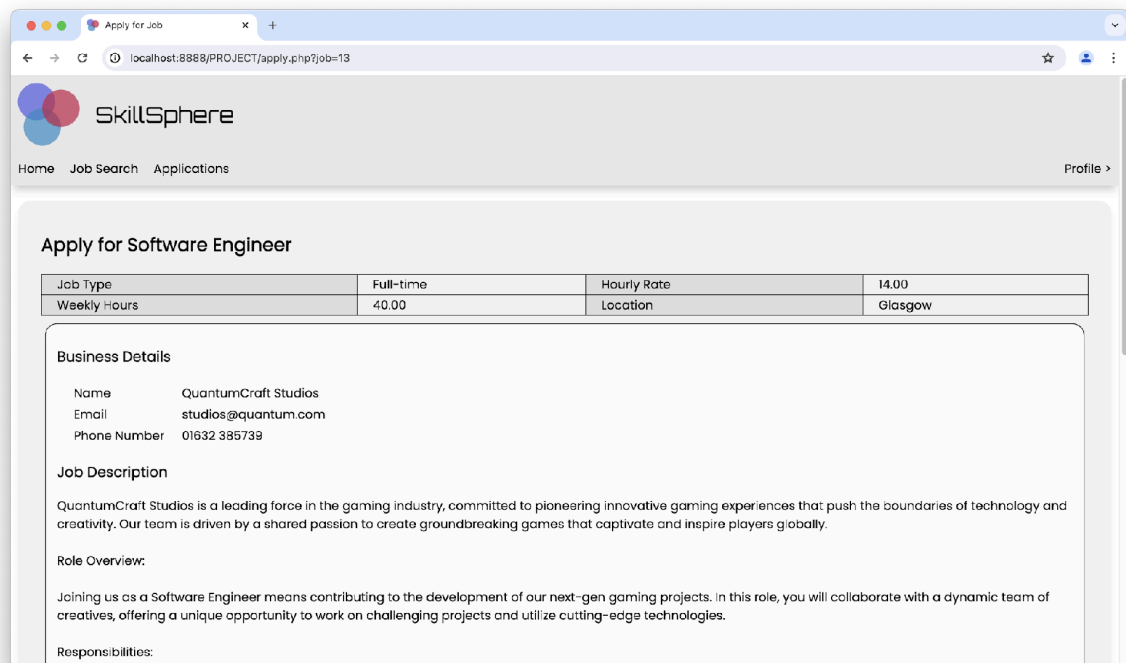
Job Search Page



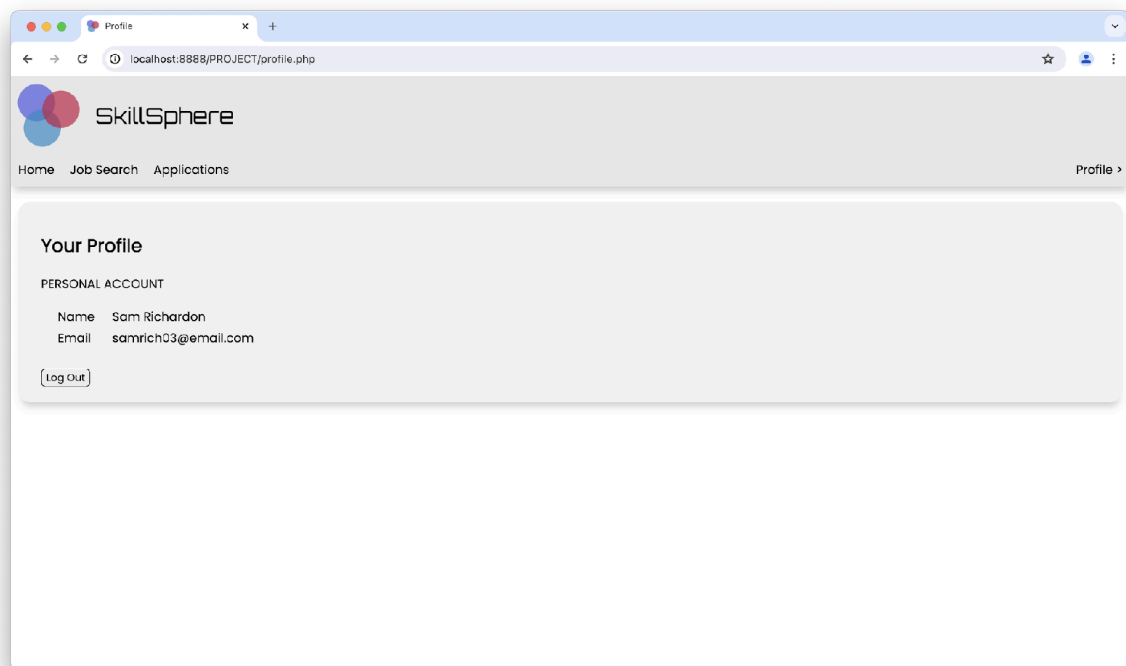
View Applications Page



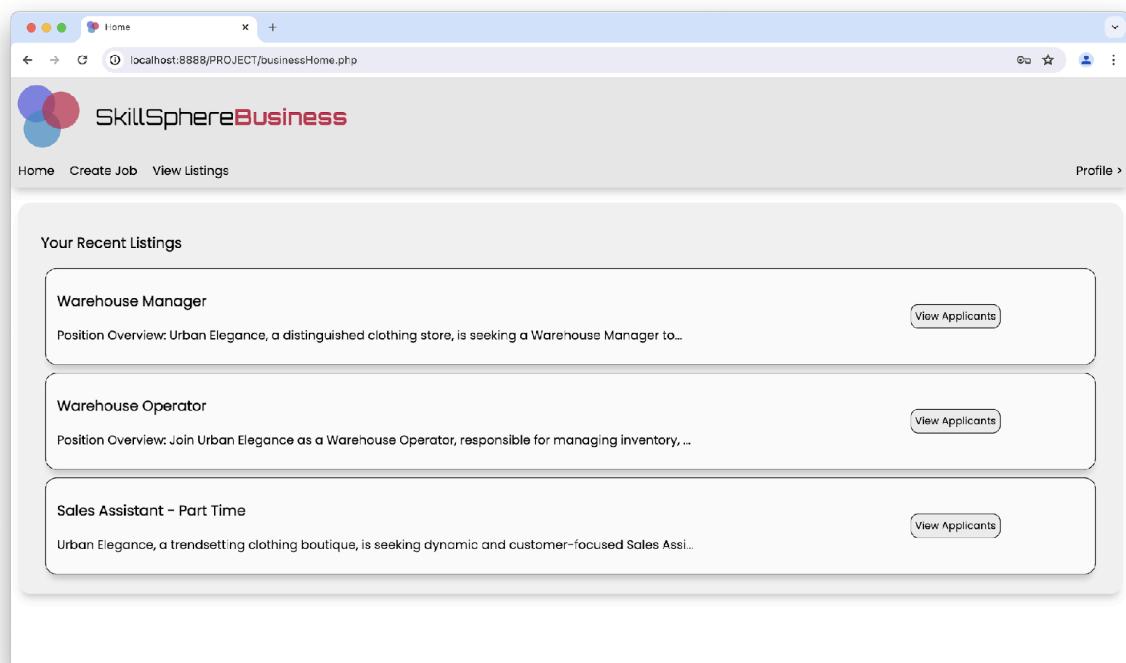
Apply for Job Page



User Profile Page



Business Home Page



Create a Job Page

The screenshot shows a web browser window with the URL `localhost:8888/PROJECT/createJob.php`. The page header features the SkillSphereBusiness logo and navigation links: Home, Create Job, View Listings, and a Profile link. The main content area is titled 'Create a Job' and contains a form with the following fields:

- Title**: A text input field.
- Job Type**: A dropdown menu with 'Part-time' selected.
- Hourly Rate**: A text input field.
- Weekly Hours**: A text input field.
- Location**: A text input field.
- Description**: A large text area for the job description.

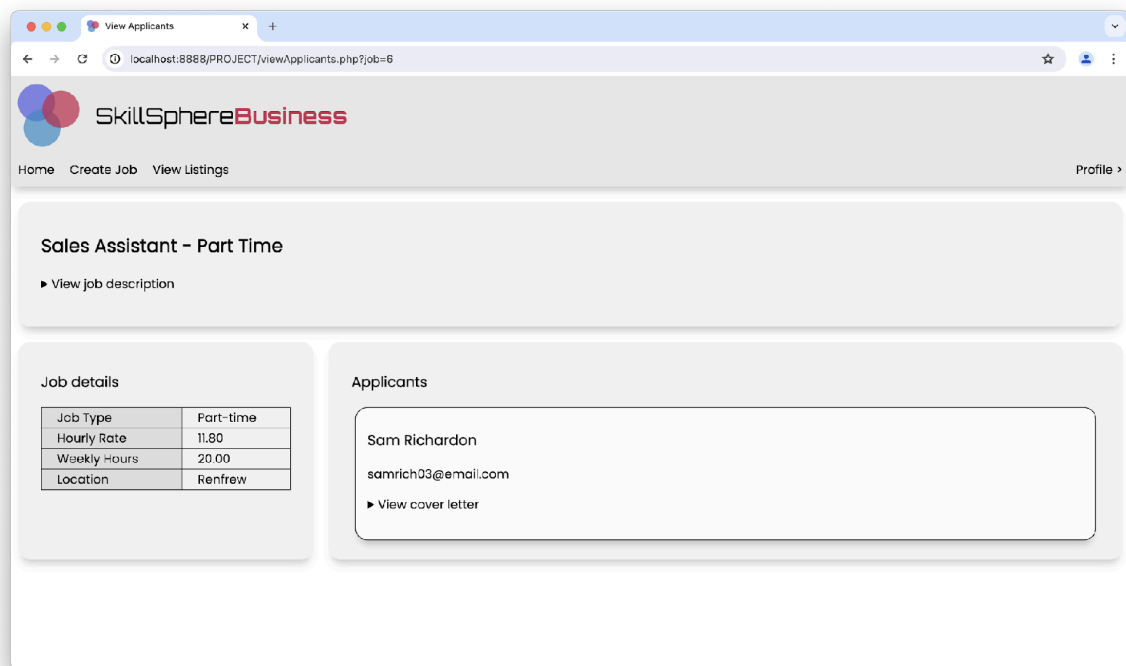
A 'Create' button is located at the bottom right of the form. A note indicates '4000 Characters Remaining' for the description field.

View Listings Page

The screenshot shows a web browser window with the URL `localhost:8888/PROJECT/viewListings.php`. The page header is identical to the 'Create a Job' page. The main content area is titled 'Your Listings' and displays a list of job postings:

- Warehouse Manager**: Position Overview: Urban Elegance, a distinguished clothing store, is seeking a Warehouse Manager to... [View Applicants](#)
- Warehouse Operator**: Position Overview: Join Urban Elegance as a Warehouse Operator, responsible for managing inventory, ... [View Applicants](#)
- Sales Assistant - Part Time**: Urban Elegance, a trendsetting clothing boutique, is seeking dynamic and customer-focused Sales Assi... [View Applicants](#)
- Sales Assistant - Full Time**: Urban Elegance, a trendsetting clothing boutique, is seeking dynamic and customer-focused Sales Assi... [View Applicants](#)

View Applicants Page



Business Profile Page

